INCH-POUND
MIL-PRF-81757/15B(AS)
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SUPERSEDING
MIL-PRF-81757/15A(AS)
10 April 2009

PERFORMANCE SPECIFICATION SHEET

BATTERY, STORAGE, AIRCRAFT, NICKEL-CADMIUM, HIGH RATE, 24-VOLT, 25-AMPERE-HOUR, LOW-MAINTENANCE, VENTED

This specification is approved for use by the Naval Air Systems Command and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-81757.

REQUIREMENTS:

1. Part or identifying number. The part or identifying numbers (PINs) of the batteries covered by MIL-PRF-81757/15 shall be M81757/15-1, M81757/15-2, and M81757/15-3, collectively known as M81757/15-X. The M81757/15-X is a limited-repair, low-maintenance, vented battery. The main differences concern different mounting configurations, weight, capacity, and power.

1.1 Harness connector assembly identifying number. The removable harness connector assembly PIN shall be M81757/15-4. The M81757/15-4 harness connector assembly is a removable and replaceable item, but shall not be repairable. The removable harness connector assembly PIN M81757/15-4 shall be installed as an integral part of PIN M81757/15-3. PIN M81757/15-4 may be purchased as an individual item separately from the PIN M81757/15-3 battery.

1.2 Harness connector assembly markings. The harness connector assembly shall contain identification markings that show the following: military part or identifying number (PIN), national stock number (NSN), contractor's Commercial and Government Entity (CAGE) code or manufacturer's name, and contract number. The identification markings label shall be attached to the wires (W7) or (W8) of the assembly as identified on figure 10.

2. Dimensions and weight. The dimensions and weight of the battery shall be as specified in 3.10.8, as modified herein and as shown on figures 1 through 7 and figure 20 contained in this specification sheet. The exterior dimensions of each cell container shall be identical to the exterior dimensions of PIN M81757/1-4.

3. Capacity. The initial capacity of the M81757/15-1 and M81757/15-2 batteries shall be not less than 27.5 ampere-hours (Ah). The initial capacity of the M81757/15-3 battery shall be not

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less than 29 ampere-hours (Ah). The minimum capacity of the M81757/15-X battery shall be not less than 25 Ah. See 3.10.14 as modified herein.

4. Interface requirements between battery and test fixtures. The battery shall contain 19 cells. The cells and other interior components of the battery shall be arranged and positioned as shown on figures 8 and 9 herein. The cell terminals shall be vertically and horizontally positioned so that each cell terminal shall mate with its matching electrical connection of the cell scanner fixture, PIN 371AS264 and the cell equalization fixture, PIN 371AS265. The fasteners specified on figures 8 and 9 of this specification sheet and 3.7.9 of MIL-PRF-81757 are required due to interfaces with the MS3509 receptacle and the cell discharge fixture.

5. Battery partition. The battery partition that contains the center row of cells and interfaces with the exterior side wall of cell number 16 (counting from the battery's negative terminal) shall be constructed of thermally conductive material. The height of the partition shall be not less than the height of the plates within cell number 16. The partition's surface adjacent to the side wall of cell number 16 shall be smooth and shall contact the cell's side wall. Spacers or air passages shall not be present at this interface. The partition shall have rigidity and strength to contain cell number 16 for the life of the battery without warping or breaking. The partition shall not distort due to undergoing any or all of the tests in MIL-PRF-81757 or MIL-PRF-81757/15.

REQUIREMENTS

6. MIL-PRF-81757 variance. The battery shall comply with MIL-PRF-81757 except as follows.

6.1 Modify the following paragraph:

3.7.2 Battery container and cover. Delete the entire text and substitute: "The battery container, mounting flanges (M81757/15-3 only), and cover shall be made of a material that is impervious to salt spray and to an aqueous solution of potassium hydroxide of 1.30 ± 0.04 specific gravity (see 6.10.1 of MIL-PRF-81757) and an alternate electrolyte if used. The container and cover may be constructed of materials that are not identical. If the cover material is electrically conductive, it shall be protected with an electrically insulating material (such as paint, epoxy, or powder coating). The protective material shall be impervious to salt spray and to an aqueous solution of potassium hydroxide of 1.30 ±0.04 specific gravity (see 3.8.1, as modified herein and 6.10.1 of MIL-PRF-81757) and an alternate electrolyte if used. Such protective materials shall not be applied to the battery container or flanges except as indicated in 3.8.1, as modified herein. The battery container, flanges, and cover shall be free of pits, blow holes, rough spots, and other deformations. The dimensions and locations of receptacles, hold-downs, latches (catches and strikes), vent tubes, and battery identification shall conform to figures 1 through 6 of this specification sheet. The cover shall be fitted with a hold-down pad

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that functions as the cover's gasket and shall have integral projections that hold down each battery cell. The cover gasket/cell hold-down pad shall conform to figure 7 of this specification sheet. The cover pad/gasket shall be sized and fitted to match the rim of the container body for a gastight fit and effective fluid seal between the cover and the container body. Metal or wood products (such as cork) shall not be used as the gasket. If elastomers are used as the gasket, the elastomer shall conform to the requirements of 3.10.3 of MIL-PRF-81757 (see 6.10.2 of MIL-PRF-81757)."

6.2 Add the following paragraphs:

"3.7.2.1 Hold-down pad (PIN M81757/15-1 only). The location of the projections shall be determined by the location of either the intercell connectors or the vented filler caps. The projections shall either contact each intercell connector or cover the entire top surface of each vented filler cap. The projections shall apply enough force on each contacted item to restrain vertical movement of the cells during vibration in accordance with 4.5.16 of MIL-PRF-81757 and 4.5.33 as specified herein. If the battery cover is made from an electrically conductive material, then a rigid insulating liner 0.062 ± 0.007 inch thick shall be placed between the inside top surface of the container cover and the cover pad/gasket. The liner shall provide complete coverage of the inside top surface of the container cover and shall conform to the requirements of 3.10.3 or 3.10.4, both of MIL-PRF-81757 as applicable (see 6.10.2 of MIL-PRF-81757)."

"3.7.2.2 Hold-down pad (PINs M81757/15-2 and M81757/15-3 only). The location of the projections shall be determined by the location of each vented filler cap. The projections shall contact each vented filler cap and shall apply enough force on each vented filler cap to restrain vertical movement of the cells during vibration in accordance with 4.5.16 of MIL-PRF-81757 (M81757/15-2 only) or both 4.5.16, as specified in MIL-PRF-81757 and 4.5.33, as specified herein (M81757/15-3 only)."

"3.7.2.3 Mounting flanges (PIN M81757/15-3 only). The dimensions and locations of the mounting flanges shall conform to figure 3 of this specification sheet. The flanges shall be attached to the battery by a non-removable means. Fasteners or adhesives shall not be used to attach the flanges to the battery container. If vertical triangular gussets (perpendicular to the horizontal and vertical members of the flange) are used to strengthen the front flange, they shall be not less than 0.50 inch from the edge of the slots in the flange. If the battery's weight is to be minimized by inserting holes in the flanges, the holes may be inserted in either flange but shall be in the vertical member only. The container and flange shall be constructed of materials that are identical. Flanges shall be free of any coating and shall meet the requirements of 3.9, 3.10.2, both of MIL-PRF-81757 and 3.10.25, as modified herein."

6.3 Modify the following paragraphs:

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3.7.5 Handles. Not applicable to M81757/15-1 and M81757/15-2. For M81757/15-3, delete the title and text and substitute: "Carrying strap. The battery case shall have two flexible carrying straps located as shown on figure 3 of this specification sheet. The straps shall be a dark color. The straps shall have a mechanism by which they can be connected to each other at two separate locations equidistant from the battery's vertical centerline. When the straps are extended for carrying the battery, the two connecting points shall be between 3 to 5 inches to the battery's vertical centerline. Fasteners or adhesive methods shall not be used to attach the straps' mounting supports to the battery. When attached to the battery, the straps' mounting supports shall not exceed the latch-to-latch width of the battery. Straps shall be free of any coating and shall meet the requirements of 3.9, 3.10.3, both of MIL-PRF-81757, 3.10.10, and 3.10.25, both as modified herein."

3.7.8 Receptacles. Add the following after the last sentence: "The battery shall contain one main power receptacle; the M81757/15-3 shall also contain a sensor receptacle. For all M81757/15-X, the J1 main power receptacle shall be as shown on figure 20 of this specification sheet. The J1 main power receptacle shall be vertically positioned so that the top of the terminal block and the cell terminal contact surfaces are in the same plane. For M81757/15-3, the J2 sensor receptacle shall conform to D38999/20KC98SN."

3.7.11.5 Cell baffle. Delete the first sentence and insert: "The cell baffle shall contain an electrolyte level marker as the reference for electrolyte adjustment (see 3.10.12 of MIL-PRF-81757). This shall apply for any vent cap style used."

3.7.11.6 Vented filler cap. Following the first sentence, insert: "The vented filler cap to be installed on each cell of the battery shall be as shown on figure 20 of this specification sheet."

6.4 Add the following paragraph:

"3.7.15 Harness assembly (PIN M81757/15-3 only). Each battery shall include a harness assembly PIN M81757/15-4 that shall be configured as shown on figure 10 of this specification sheet. The schematic and pin assignments shall conform to figure 11 of this specification sheet. The harness assembly shall be replaceable. Wire-wound resistors shall not be used. All electrical connections made to the receptacle and harness devices shall be potted with a compound that will not support flame, absorb moisture, flow, crack, or shrink during any of the tests of 4.5 of MIL-PRF-81757. The harness assembly and potting compound shall meet the requirements of 3.10.3 of MIL-PRF-81757. The harness assembly shall use two temperature sensor devices (thermistors) with a resistance output conforming to figure 12 of this specification sheet. The devices shall have a tolerance of ±0.2 °C over the range 0 °C to 70 °C. The manufacturer shall provide three samples of the M81757/15-2 harness assembly in addition to the samples of 4.2.2 of MIL-PRF-81757."

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"3.7.15.1 Harness alkali resistance. External and internal metal parts of each battery, cell, and component shall not show cracking, pitting, chipping, scaling, corrosion, or other deleterious effects during or after the testing specified in 4.5.36 of MIL-PRF-81757/15. The battery shall also meet the requirements of 3.9 of MIL-PRF-81757.

6.5 Modify the following paragraphs:

3.8.1 Color. Delete the third sentence and substitute: "The battery container shall be the natural color of the container material (see 6.10.7 of MIL-PRF-81757) except as follows. The container cover shall be a blue color as specified above. The coloring of the battery identification (see 3.8.4, as modified herein) shall be either:

"a. White print on blue label backgrounds colored the same blue specified above, or

"b. Blue lettering applied directly to the container material without the use of a colored background."

3.8.3.1 Batteries. In the first sentence, change "marked on" to "marked, impressed, or embossed on".

3.8.4 Battery identification. Delete the third sentence and substitute: "The marking shall be either in white print on a blue background (see 3.8.1, as modified herein) or in light blue or medium blue print on the natural color of the container material. The blue color shall provide sufficient contrast against the natural color of the container for easy visibility under natural light."

3.8.4.1 Limited-repair batteries. In the first sentence, change "figure 1" to "figure 13 of this specification sheet".

3.8.4.2 Repairable batteries. In the first sentence, change "figure 2" to "figure 14 of this specification sheet".

3.8.4.3 Battery caution and receptacle marking. In the second and third sentences, change "figure 3" to "figure 14 of this specification sheet". Insert the following after the third sentence: "The assigned CAGE code shall be inserted at the indicated area."

3.10.8 Dimensions and weights. Delete the entire text and substitute: "The dimensions and configuration of the battery container and container cover shall be as shown on figures 1 through 7 of this specification sheet. The weight of the battery shall be not greater than that shown on figure 20 of this specification sheet with the electrolyte level adjusted in accordance with 4.4.4 of MIL-PRF-81757. The base dimensions of the container shall be within specified tolerance for a distance of not less than 1.8 inches up from the bottom of the container.

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Beyond this distance, the case dimensions may exceed the base dimension by 50 percent of the specified tolerance."

6.6 Add the following paragraphs:

"3.10.8.1 PIN M81757/15-1 only. All dimensions are in inches and are prior to coating except overall width (10.148 max), overall length (10.662 max), overall height (8.812 max), container base width (7.838 + 0.062, -0.15), and container base length (9.938 ± 0.062). Coating shall not be applied to the battery container but may be applied to the battery cover."

"3.10.8.2 PIN M81757/15-2 only. All dimensions are in inches and are prior to coating except overall width (9.998 max), overall length (10.662 max), overall height (8.812 max), container base width (7.688 ±0.062), and container base length (9.938 ±0.062). Coating shall not be applied to the battery container but may be applied to the battery cover."

"3.10.8.3 PIN M81757/15-3 only. All dimensions are in inches and are prior to coating except overall width (9.998 max), overall height (8.812 max), container base width (7.688 ±0.062), and container base length (9.938 ±0.062). The vent tube on the front face of the battery shall not be included in the overall length dimension. Coating shall not be applied to the battery container or flanges but may be applied to the battery cover."

6.7 Modify the following paragraphs:

3.10.10 Handle strength. Not applicable to M81757/15-1 and M81757/15-2. For M81757/15-3, delete the title and text and substitute: "Carrying strap dimensions and strength. The locations of the carrying straps shall conform to figure 3 of this specification sheet. Each of the battery's carrying straps shall be not less than 1 inch (nominal) wide. The straps shall be long enough to avoid blocking the J1 and J2 receptacles during battery maintenance. Each strap shall be long enough to permit the easy movement of the test box of 4.5.9, as modified herein between the strap under test and the battery. During this movement, the box shall not contact the vent tube on the battery lid. The straps shall not show any evidence of breaking away from the container or other damage after being tested in accordance with 4.5.9, as modified herein. The straps and battery shall also meet the requirements of 3.9 of MIL-PRF-81757."

3.10.14 Capacity. Delete the first sentence and insert: "The initial capacity of the M81757/15-1 and M81757/15-2 batteries shall be not less than 27.5 ampere-hours (Ah). The initial capacity of the M81757/15-3 battery shall be not less than 29 ampere-hours (Ah). The minimum capacity of the M81757/15-X battery shall be not less than 25 Ah."

3.10.14.1 Initial capacity. Delete the text and substitute: "To meet the initial capacity requirement of the M81757/15-1 and M81757/15-2 batteries, each cell shall produce a

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potential of not less than 1.0 volt at not less than 60 minutes into the constant 25-ampere discharge after being tested in accordance with 4.5.13 of MIL-PRF-81757 at room temperature (see 4.4.1 of MIL-PRF-81757). To meet the initial capacity requirement of the M81757/15-3 battery, each battery shall produce a potential of not less than 18 volt at not less than 70 minutes into the constant 25-ampere discharge after being tested in accordance with 4.5.13 at room temperature (see 4.4.1 of MIL-PRF-81757).""

3.10.15 Humidity and charge retention. In step 3.10.15.a(1) of MIL-PRF-81757, delete "60 minutes" and replace with "48 minutes".

3.10.22 Constant voltage discharge. Delete the text and substitute: "The battery shall produce not less than the current specified below after being tested in accordance with 4.5.22 of MIL-PRF-81757. The battery shall also meet the requirements of 3.9 of MIL-PRF-81757 during and after charge and discharge."

Battery PIN	Temperature	Current at 15 seconds	Average current over the total 60 seconds
M81757/15-1 and M81757/15-2	24.0 ±2.0 ºC	700 A	650 A
M81757/15-1 and M81757/15-2	-26.0 ±2.0 ºC	350 A	350 A
M81757/15-3	24.0 ±2.0 ºC	630 A	585 A
M81757/15-3	-26.0 ±2.0 ºC	315 A	315 A

3.10.23 Cycling. Delete the text of 3.10.23.c of MIL-PRF-81757 and insert: "c. After successfully completing the required number of duty cycles, the battery's electrical capacity shall comply with the following:

"(1) Qualification inspection

"(a) M81757/15-1 and M81757/15-2: The battery shall deliver not less than 20 Ah when subjected to 4.5.23.d, as modified herein. To comply with the 20-Ah requirement, the battery shall produce a potential of not less than 18.0 volts (end voltage) at not less than 60 minutes into the constant 20-ampere discharge.

"(b) M81757/15-3: The battery shall deliver not less than 25 Ah when subjected to 4.5.23.d, as modified herein. To comply with the 25-Ah requirement, the battery shall produce a potential of not less than 18.0 volts (end voltage) at not less than 60 minutes into the constant 25-ampere discharge.

"(2) Group C inspection: The battery shall deliver not less than 20 Ah when subjected to 4.5.23.d, as modified herein. To comply with the 20-Ah requirement, the battery

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shall produce a potential of not less than 18.0 volts (end voltage) at not less than 48 minutes into the constant 25-ampere discharge."

3.10.23 Cycling. Delete the text of 3.10.23.d and insert: "d. The battery shall successfully complete 700 duty cycles and 200 duty cycles for qualification inspection and 200 duty cycles for group C inspection when subjected to 4.5.23, as modified herein."

3.10.25 Physical integrity at high temperature. For M81757/15-1 and M81757/15-2, in the first sentence, delete "and handles". For M81757/15-3, in the first sentence, replace "vent tubes and handles" with "vent tubes, flanges, and straps".

6.8 Add the following paragraphs:

"3.10.31 Slope discharge (PIN M81757/15-1 only). The battery shall perform as follows after being tested in accordance with 4.5.32 as specified herein..

"a. The battery shall produce a potential of not less than 12.0 volts after being discharged in accordance with figure 16 of this specification sheet for two consecutive discharges.

"b. The battery shall also meet the requirements of 3.9 of MIL-PRF-81757 during and after testing and discharge."

"3.10.32 Low frequency vibration (PIN M81757/15-1 only). The battery shall perform as follows after being tested in accordance with 4.5.33 as specified herein.

"a. During the charging of the battery in the test of 4.5.33 as specified herein, the increase in the charge current shall be not greater than 5 amperes from the lowest value reached.

"b. After being tested in accordance with 4.5.33, the battery shall produce a potential of not less than 12.0 volts when discharged in accordance with figure 17 of this specification sheet during each of the 17 vibration cycles.

"c. The battery shall also meet the requirements of 3.9 of MIL-PRF-81757 during and after testing and discharge."

"3.10.33 APU discharge (PIN M81757/15-3 only). The battery shall perform as follows after being tested in accordance with 4.5.35 of MIL-PRF-81757/15.

"a. The battery shall produce a potential of not less than 14.0 volts during the three APU start profiles (60-67 sec, 127-134 sec, and 194-201 sec) of figure 17 of MIL-PRF-81757/15, and not less than 18.0 volts during all other discharge periods.

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"b. The battery shall also meet the requirements of 3.9 of MIL-PRF-81757 during and after testing and discharge."

“c. The battery shall provide not less than 8 minutes of total discharge time during the APU discharge profile test.

d. The measured thermistor resistance values shall be within 5% of the values in Figure 12 for the temperature at which the battery has been stabilized.

VERIFICATION

7. MIL-PRF-81757 variance. The battery shall comply with MIL-PRF-81757 except as follows.

7.1 Modify the following paragraph:

4.3.3 Sample size and selection for groups A, B, and C inspection. Insert after the last sentence: "Samples for group B subgroup I shall be selected from units that have passed group A inspection. Samples for group B subgroup II shall be selected from the samples that were subjected to group B subgroup I inspection. After selection of subgroup II samples, the remainder of group B subgroup I units may then be released for shipment."

7.2 Add the following paragraphs:

"4.4.1.1 Temperature measurement location. When the battery temperature is to be measured during laboratory testing, the temperature shall be measured at the location between the two cells shown on figures 8 (for PINs M81757/15-1 and M81757/15-2) and 9 (M81757/15-3) of this specification sheet. The temperature shall be measured with a temperature sensor (thermocouple or thermistor) inserted between the two cells and positioned in the approximate center of the outer sidewall of each cell equidistant from top to bottom and side to side."

"4.4.7 Mounting for vibration testing (PIN M81757/15-1 only). The battery shall be clamped in the cradle by two flat bars, with one of the bars bearing on the cover adjacent to each of the left and right edges of the cover, and extending inward from the edges 0.844 ± 0.100 inch from the sides, measured in the lower 2/10 of the height, and extending from the front edge of the cover to the back edge of the cover. The hold-down bars shall be clamped to the cradle by four vertical 5/16-24 threaded rods with nuts tightened to 45 ±3 inch-pounds, bearing on the hold-down bars through cross-beams. The rods shall be arranged symmetrically about the battery, with two rods on each side of the battery. Two of the rods shall be 2.500 ±0.100 inches from the front of the battery, and two of the rods shall

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be 2.500 ±0.100 inches from the back of the battery, measured in the lower 2/10 of the height of the battery."

7.3 Modify the following paragraphs:

4.5.8 Dimensions & Weight. Add: Measure the resistance of the two thermistors on pins C-F and H-A. Values must be within 5% of those seen on figure 12.

4.5.9 Handle strength test. Not applicable to M81757/15-1 and M81757/15-2. For M81757/15-3, delete the title and text and substitute: "Carrying strap dimensions and strength test. Subject each strap individually to the following tests. Measure the width of each strap. Pass a box of dimensions 5.25 inches wide by 3 inches high by 6 inches long in between the battery and each strap. The box shall be oriented such that the 6-inch dimension is parallel to the length of the battery (front to back) and the 3-inch dimension is vertical. Remove the box. Swing each strap out from the container against its stop. Apply a force of 75 pounds (333.6 Newtons) at the grasp of each strap. Apply the force in a direction of maximum torque acting on the strap stop. Apply the force continuously for not less than 5 minutes. Examine each strap and the battery for the requirements of 3.10.10, as modified herein."

4.5.23 Cycling test. Insert the following after the last sentence: "The battery shall undergo the 700 duty cycles test and the 200 duty cycles test for qualification inspection and only the 200 duty cycles test for group C inspection. The temperature near the center of the battery shall be monitored and recorded throughout this test."

4.5.23 Cycling test. Delete the text of 4.5.23.b(1) and insert: "(1) Discharge the M81757/15-1 and M81757/15-2 batteries for 20 seconds through a 30-milliohm resistance for qualification inspection. Discharge the M81757/15-3 battery for 20 seconds at a constant 270 amperes for qualification inspection. Discharge the M81757/15-X battery for 20 seconds through a 24-milliohm resistance for qualification inspection and group C inspection."

4.5.23 Cycling test. Delete the text of 4.5.23.b(4)(a) and insert: "For low-maintenance vented batteries, charge at 28.25 ±0.1 volts for 120 minutes."

4.5.23 Cycling test. In the first sentence of 4.5.23.d, delete "at the C-rate" and insert a new second sentence: "The M81757/15-1 and M81757/15-2 batteries shall be discharged at a constant 20 amperes during qualification inspection. The M81757/15-3 battery shall be discharged at a constant 25 amperes during qualification inspection. The M81757/15-X battery shall be discharged at a constant 25 amperes during group C inspection."

4.5.25 Physical integrity at high temperature. For M81757/15-1 and M81757/15-2, in step c, delete the second sentence. For M81757/15-3, in the second sentence of step c, change "handles" to "straps". For M81757/15-3, delete the last sentence and insert "Observe any

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evidence of shearing, breaking, bending, or deterioration at the point of connection of the vent tube and the battery, mounting flanges and the battery, or other distortion of the case. When at high temperature, measure the resistance of the thermistors. These values must be within 5% of the values shown in Figure 12."

7.4 Add the following paragraphs:

"4.5.32 Slope discharge test (PIN M81757/15-1 only). This test shall consist of the following steps.

"a. Place the fully charged battery in a 38° ±2 °C (100° ±5 °F) environment and allow its temperature to stabilize.

"b. While the battery remains within the temperature environment of 4.5.32.a, discharge the battery in accordance with figure 16 of this specification sheet.

"c. Place the battery in a room temperature environment and constant potential charge the battery at 27.74 ±0.10 volts for 5.0 ±0.1 hours.

"d. Repeat 4.5.32.a and 4.5.32.b.

"e. Continuously measure the terminal voltage and temperature of the battery.

"f. Examine the battery for the requirements of 3.10.31 as specified herein."

"4.5.33 Low frequency vibration test (PIN M81757/15-1 only). Subject the battery to seventeen 5-hour cycles of vibration along its vertical axis. The battery shall not receive any maintenance throughout this test. If the test battery's terminals are an open circuit for not less than 24 hours after starting and before completing this test, the battery shall be constant potential charged in accordance with 4.4.2 of MIL-PRF-81757 before continuing the test. This test shall consist of the following steps.

"a. Remove the cover from the battery.

"b. Constant potential charge the battery in accordance with 4.4.2.

"c. Vibration cycle: The following steps comprise one vibration cycle. Steps 4.5.33.c(3) and 4.5.33.c(4) will amount to a total of 5 hours of vibration.

"(1) Place the battery in a 38° ±2 °C (100° ±5 °F) environment and allow its temperature to stabilize. Perform the discharge of figure 16 of this specification sheet.

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"(2) Remove the battery from the 38° ±2 °C (100° ±5 °F) environment, secure the cover onto the battery, and proceed immediately with 4.5.33.c(3).

"(3) Attach the battery (oriented in its normal upright position) to the vibration equipment (see 4.4.7 as specified herein). Ventilate the battery through the vent tubes at 1 ±10 percent cubic feet per minute during vibration. Vibrate the battery along its vertical axis in accordance with figure 18 of this specification sheet. Constant potential charge the battery at 27.74 ±0.10 volts while the battery is subjected to vibration.

"(4) Repeat 4.5.33.c(3) fourteen times.

"d. Repeat 4.5.33.c sixteen times. The seventeen vibration cycles will amount to a total of 85 hours of vibration. Step 4.5.33.c(3) shall be repeated not more than two times within a 24-hour period.

"e. Continuously measure the battery's terminal voltage and temperature and the charging current.

"f. Examine the battery for the requirements of 3.9 of MIL-PRF-81757 and 3.10.32 as specified herein."

"4.5.34 Gunfire vibration test (PIN M81757/15-3 only). Perform the test of 4.5.16 of MIL-PRF-81757 with the following modification. Delete the sixth sentence and insert: 'Vibrate the battery in accordance with figure 19 of this specification sheet along each of three mutually perpendicular axes.' The discrete frequency sinusoidal excitation defined in table III of figure 19 of this specification sheet shall be combined with the narrow band random vibration of table II of figure 19 superimposed over the broadband random background of table I of figure 19."

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"4.5.35 APU discharge test (PIN M81757/15-3 only). This test shall consist of the following steps.

"a. Constant current charge the battery in accordance with 4.4.3.

"b. Place the fully charged battery in a -17.7 ºC ±2 ºC (-0 ºF ±3.6 ºF) environment and allow its temperature to stabilize.

"c. While the battery remains within the temperature environment of 4.5.35.b, measure the thermistor resistance on pins C-F and H-A. These values must be within 5 percent of the values shown on Figure 12. Proceed to discharge the battery in accordance with figure 17 of MIL-PRF-81757/15. Continue the discharge at the final current level of 100 amps until terminal voltage reaches 18.0 volts.

"d. Stabilize the battery at room temperature in accordance with 4.4.1. Constant potential charge the battery in accordance with 4.4.2.

"e. Continuously measure the terminal voltage and temperature of the battery.

"f. Examine the battery for the requirements of 3.10.33 of MIL-PRF-81757/15. Stabilize the battery at room temperature in accordance with 4.4.1."

"4.5.36 Harness alkali resistance test. Obtain an M81757/15-4 sensor/harness assembly. Do not disassemble the M81757/15-4 assembly. The test shall consist of the following steps.

a.Install the harness connector assembly on the inside of a container which is capable of withstanding 1.300 ±0.025 specific gravity potassium hydroxide electrolyte.

b. Attach the harness connector to the container in the same way it is attached to the battery. Fill the container with electrolyte solution of potassium hydroxide of 1.30 ±0.04 specific gravity to cover all parts of the harness. Cover the container and place the container in a 66° ±2 °C (150° ±3.6 °F) environment to bring the electrolyte solution to 150 ºF.

c. Soak the harness in the electrolyte at 150 ºF for 168.0 ±1.0 hours. At the end of the soak period, remove the harness assembly from the container. Rinse the assembly in water and wipe dry.

d. Measure the thermistor resistance on pins C-F and H-A. These values must be within 5 percent of the values shown on figure 12. Examine the assembly for the requirements of 3.7.15.1 of MIL-PRF-81757/15.

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7.5 Modify the following paragraphs, tables, and figures:

6.1 Intended use. Add the following after the last sentence: "The M81757/15-X limited-repair battery is designed so that its cells and all components associated with installing the cells cannot be removed and replaced. The battery should be requisitioned and used only by those DoD Departments and Agencies whose aircraft battery maintenance/repair policies, procedures, and practices disallow the battery's cells to be removed and replaced."

6.10.3 Handles. Not applicable.

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TABLE III. Qualification inspection of batteries. Delete and substitute the following:
TABLE III. Qualification inspection of batteries.

Test number	Examinations and tests	Sample number					Requirement paragraph	Test paragraph
		1	2	3	4	5		
1	INSPECTION FOR DAMAGE	X	X	X	X		3.10.1 (G)	4.5.1 (G)
2	VISUAL AND MECHANICAL	X	X	X	X		3.10.6 (G), Req 4 (N), and 5 (N)	4.5.6 (G)
3	CELL INSERTION FORCE TEST (REPAIRABLE BATTERIES ONLY)	X	X	X	X		3.10.7 (G)	4.5.7 (G)
4	DIMENSIONS AND WEIGHT	X		X			3.10.8 (S), 3.10.9 (G), Req. 2 (N)	4.5.8 (G)
5	CARRYING STRAP STRENGTH TEST (FOR PIN M81757/15-3 ONLY)			X	X		3.9 (G), 3.10.10 (S)	4.5.9 (S)
6	COVER SEAL TEST			X	X		3.9 (G), 3.10.11 (G)	4.5.10 (G)
7	CELL VENT TEST	X	X	X	X		3.10.13 (G)	4.5.12 (G)
8	INITIAL CAPACITY DISCHARGE	X	X	X	X		3.10.14 (S), Req. 3 (N)	4.5.13 (G)
9	HUMIDITY AND CHARGE RETENTION			X	X		3.9 (G), 3.10.15 (S) Req. 3 (N)	4.5.14 (G)
10	SHOCK TEST (BASIC DESIGN)			X	X		3.9 (G), 3.10.16 (G)	4.5.15 (G)
11	VIBRATION TEST			X	X		3.9 (G)	4.5.16 (G)
12	GUN FIRE VIBRATION TEST (PIN M81757/15-3 ONLY)			X	X		3.9 (G)	4.5.34 (N)
13	TEMPERATURE SHOCK TEST			X	X		3.9 (G), 3.10.17 (G)	4.5.17 (G)
14	ALTITUDE TEST			X	X		3.9 (G), 3.10.18 (G)	4.5.18 (G)
15	SALT FOG TEST	X	X				3.9 (G), 3.10.19 (G)	4.5.19 (G)

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TABLE III. Qualification inspection of batteries - Continued.

Test number	Examinations and tests	Sample number					Requirement paragraph	Test paragraph
		1	2	3	4	5		
16	20-SECOND PULSE DISCHARGE			X	X		3.9 (G), 3.10.20 (G)	4.5.20 (G)
17	MEDIUM-RATE DISCHARGE OPERATING POSITION TEST			X	X		3.9 (G), 3.10.21 (G)	4.5.21 (G)
18	CONSTANT VOLTAGE DISCHARGE	X	X	X	X		3.9 (G), 3.10.22 (S)	4.5.22 (G)
19	CYCLING TEST 2/	X	X	X	X		3.9 (G), 3.10.23 (S)	4.5.23 (S)
20	TEMPERATURE RISE AND FLOAT	X	X				3.9 (G), 3.10.9 (G), 3.10.24 (G)	4.5.24 (G)
21	APU DISCHARGE TEST (PIN M81757/15-3 ONLY)	X	X				3.9 (G), 3.10.33 (N)	4.5.35 (N)
22	PHYSICAL INTEGRITY AT HIGH TEMPERATURE (85ºC)	X					3.9 (G), 3.10.25 (S)	4.5.25 (S)
23	CONSTANT VOLTAGE DISCHARGE		X				3.9 (G), 3.10.22 (S)	4.5.22 (G)
24	INTERNAL PRESSURE TEST	X					3.9 (G), 3.10.26 (G)	4.5.26 (G)
25	DIELECTRIC STRENGTH AND INSULATION RESISTANCE TEST	X	X	X	X		3.10.27 (G)	4.5.27 (G)
26	GAS BARRIER MATERIAL TEST (ONLY FOR BATTERIES WITH GAS BARRIER MATERIALS OTHER THAN THOSE LISTED IN 6.10.4 (G))		X	X			3.10.28 (G)	4.5.28 (G)
27	STORAGE EFFECTS					X	3.9 (G), 3.10.29 (G) 3.10.24(G)	4.5.29 (G)
28	SHELF LIFE	X			X		3.9 (G), 3.10.30 (G)	4.5.30 (G)
29	LOW FREQUENCY VIBRATION TEST (PIN M81757/15-1 ONLY)	X	X	X	X		3.9 (G), 3.10.32 (N)	4.5.33 (N)

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TABLE III. Qualification inspection of batteries - Continued.

Test number	Examinations and tests	Sample number	Requirement paragraph	Test paragraph
30	CELL BAFFLE TEST	SAMPLE CELL OF 4.2.1 (G)	3.7.11.5 (S), 3.10.12 (G)	4.5.11 (G)
31	TAMPER-RESISTANT FASTENER TEST FOR FASTENERS OTHER THAN SHEAR-HEAD SCREWS	SAMPLE CELL OF 4.2.1 (G)	3.7.9 (G)	4.5.31 (G)
32	SOAK TEST	SAMPLE HARNESS ASSEMBLY OF 3.7.15 (N)	3.7.15.1 (N)	4.5.36 (N)

1/ The notations in parentheses specify the source of the paragraph. "(G)" means the paragraph as shown in MIL-PRF-81757. (S)" shows that the paragraph has been modified by this specification sheet. "(N)" indicates that a new paragraph or requirement has been added by this specification sheet.

2/ For qualification testing perform both the 700 and 200 life cycle tests. The 200 life cycle test shall be performed on battery sample numbers 3 and 4 after the initial capacity discharge test (4.5.13) and before the humidity and charge retention test (4.5.14).

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TABLE VII. Group B inspection. Delete the table and insert the new table VII shown below.

TABLE VII. Group B inspection.

Test No.	Examinations and Tests	Requirement Paragraph 1/	Test Paragraph 1/
	Subgroup I:		
I-1	Dimensions and weight	3.10.8 (S), 3.10.9 (G), Req. 2 (N)	4.5.8 (G)
I-2	Cover seal test	3.9 (G), 3.10.11 (G)	4.5.10 (G)
I-3	Vented filler cap test (two vented filler caps per battery)	3.9 (G), 3.10.5 (G)	4.5.5.1 (G)
	Subgroup II:		
II-1	Initial capacity discharge	3.10.14 (S), Req. 3 (N)	4.5.13 (G)
II-2	20-second pulse discharge	3.9 (G), 3.10.20 (G)	4.5.20 (G)
II-3	Constant voltage discharge OR APU discharge test (PIN M81757/15-3 only)	3.9 (G), 3.10.22 (S) 3.9 (G), 3.10.33 (N)	4.5.22 (G) 4.5.35 (N)

1/ The notations in the parentheses specify the source of the paragraph. "(G)" means the paragraph as shown in MIL-PRF-81757. "(S)" shows that the paragraph has been modified by this specification sheet. "(N)" indicates that a new paragraph or requirement has been added by this specification sheet.

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TABLE VIII. Group C inspection. For the visual and mechanical test (test numbers I-2, II-1, and III-1), add "requirements 4 and 5" to the requirement paragraph column. For the dimensions and weight test (test number I-3), add "requirement 2" to the requirement paragraph column. For the initial capacity discharge tests (test numbers I-6, II-5, and III-4) and the humidity and charge retention test (test number II-6), add "requirement 3" to the requirement paragraph column. Add the following test:

Test No.	Examinations and tests 1/	Requirement paragraph 1/	Test paragraph 1/
	Subgroup I:		
I-6a	Slope discharge test (PIN M81757/15-1 only)	3.9 (G), 3.10.31 (N)	4.5.32 (N)
I-6a	APU discharge test (PIN M81757/15-3 only)	3.9 (G), 3.10.33 (N)	4.5.35 (N)

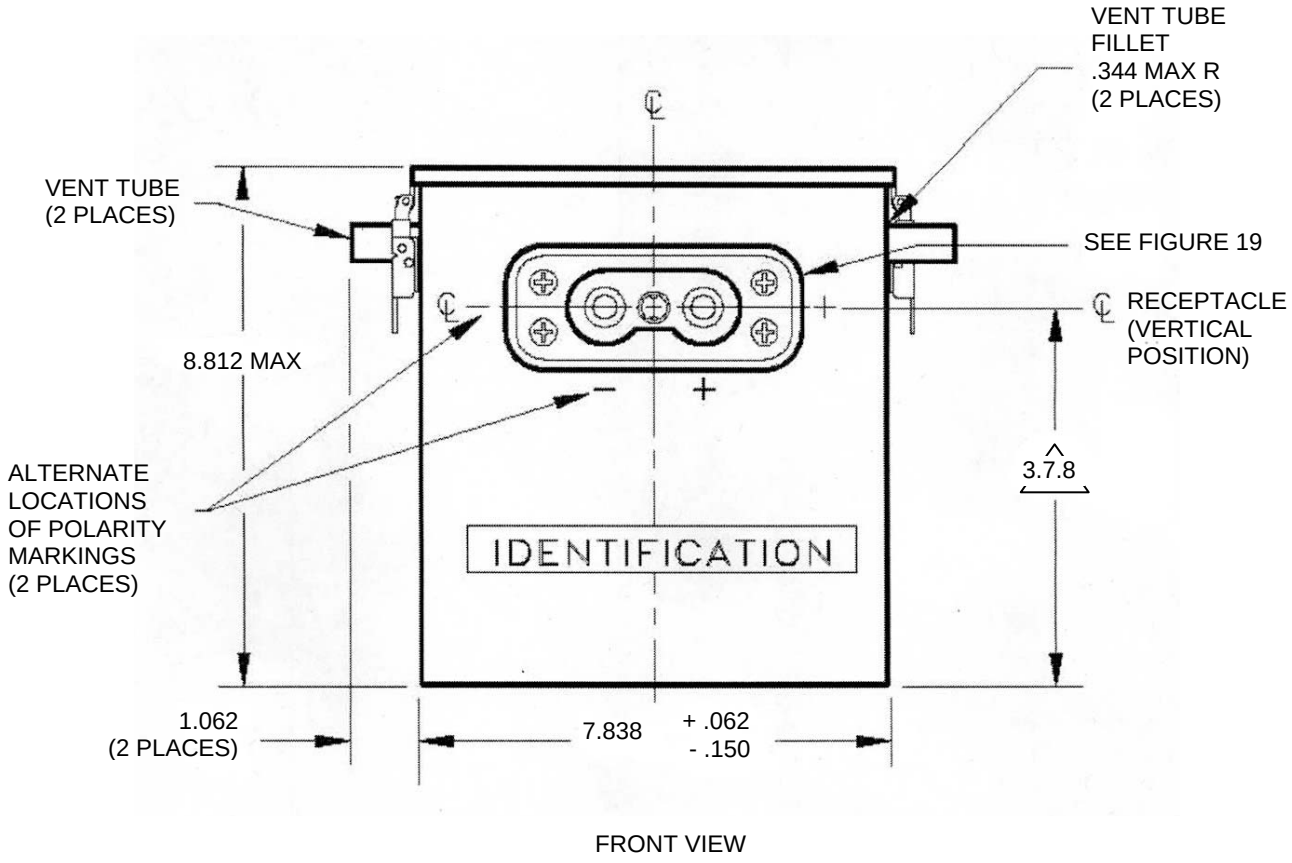
1/ The notations in the parentheses specify the source of the paragraph. "(G)" means the paragraph as shown in MIL-PRF-81757. "(N)" indicates that a new paragraph or requirement has been added by this specification sheet.

FIGURE 1. Marking for limited-repair batteries. Delete figure 1 of MIL-PRF-81757 and insert figure 13 of this specification sheet.

FIGURE 2. Marking for repairable batteries. Delete figure 2 of MIL-PRF-81757 and insert figure 14 of this specification sheet.

FIGURE 3. Marking for battery caution and receptacle. Delete figure 3 of MIL-PRF-81757 and insert figure 15 of this specification sheet.

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FRONT VIEW

NOTES:

1. ALL DIMENSIONS ARE IN INCHES.
2. ALL TOLERANCES ± .062 UNLESS OTHERWISE SPECIFIED.

FIGURE 1. Dimensions and configuration for PIN M81757/15-1.

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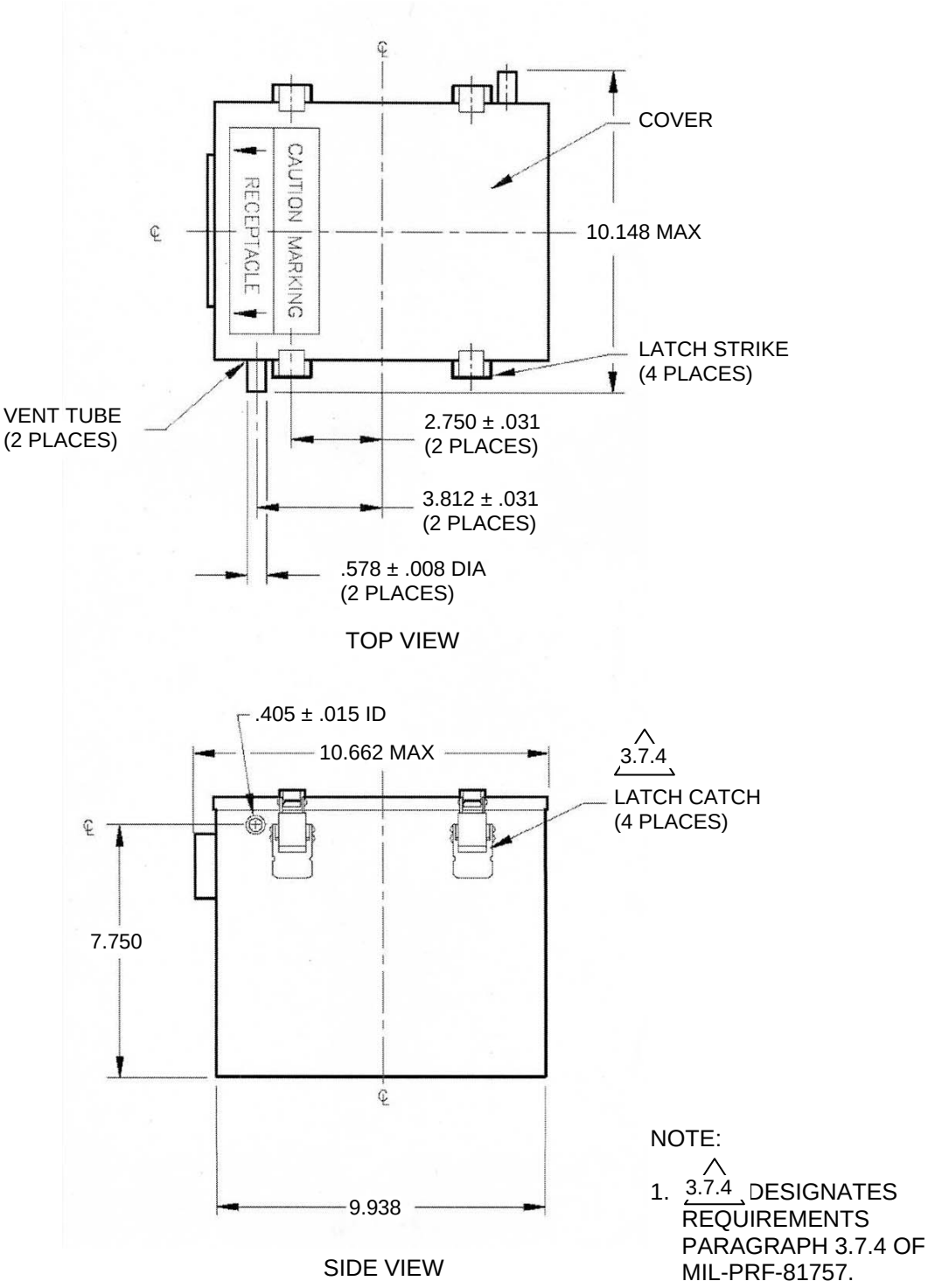
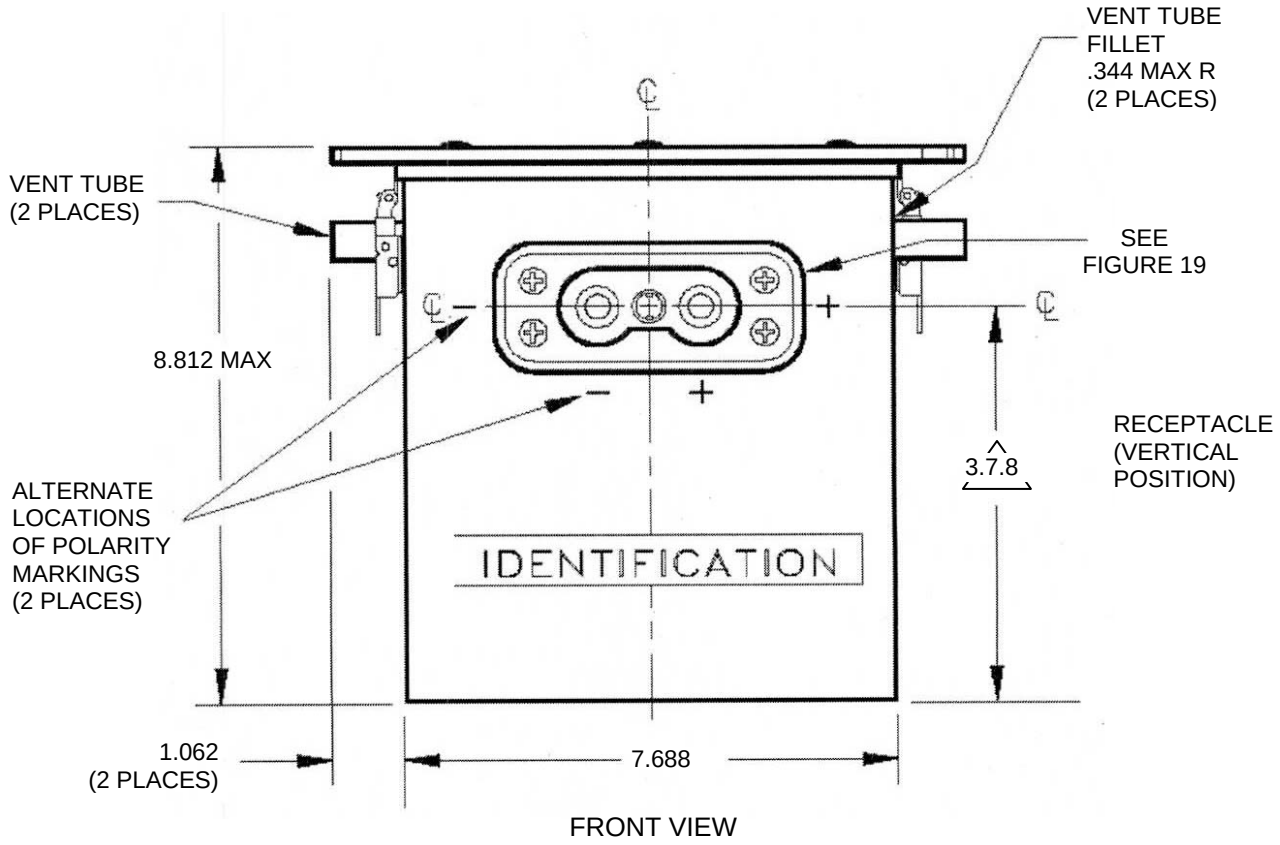


FIGURE 1. Dimensions and configuration for PIN M81757/15-1 - Continued.

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NOTES:

1. ALL DIMENSIONS ARE IN INCHES.
2. ALL TOLERANCES ± .062 UNLESS OTHERWISE SPECIFIED.

FIGURE 2. Dimensions and configuration for PIN M81757/15-2.

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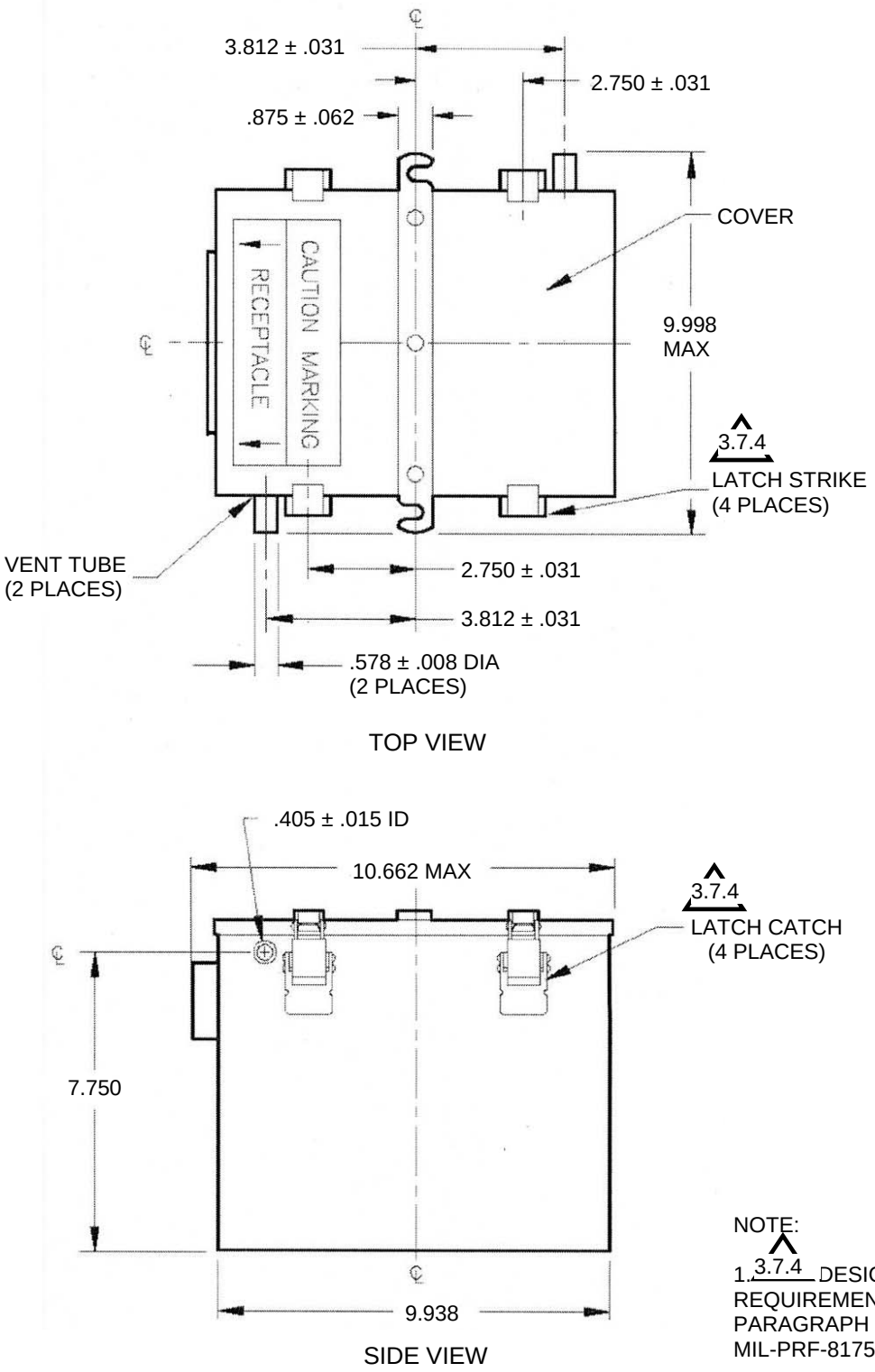
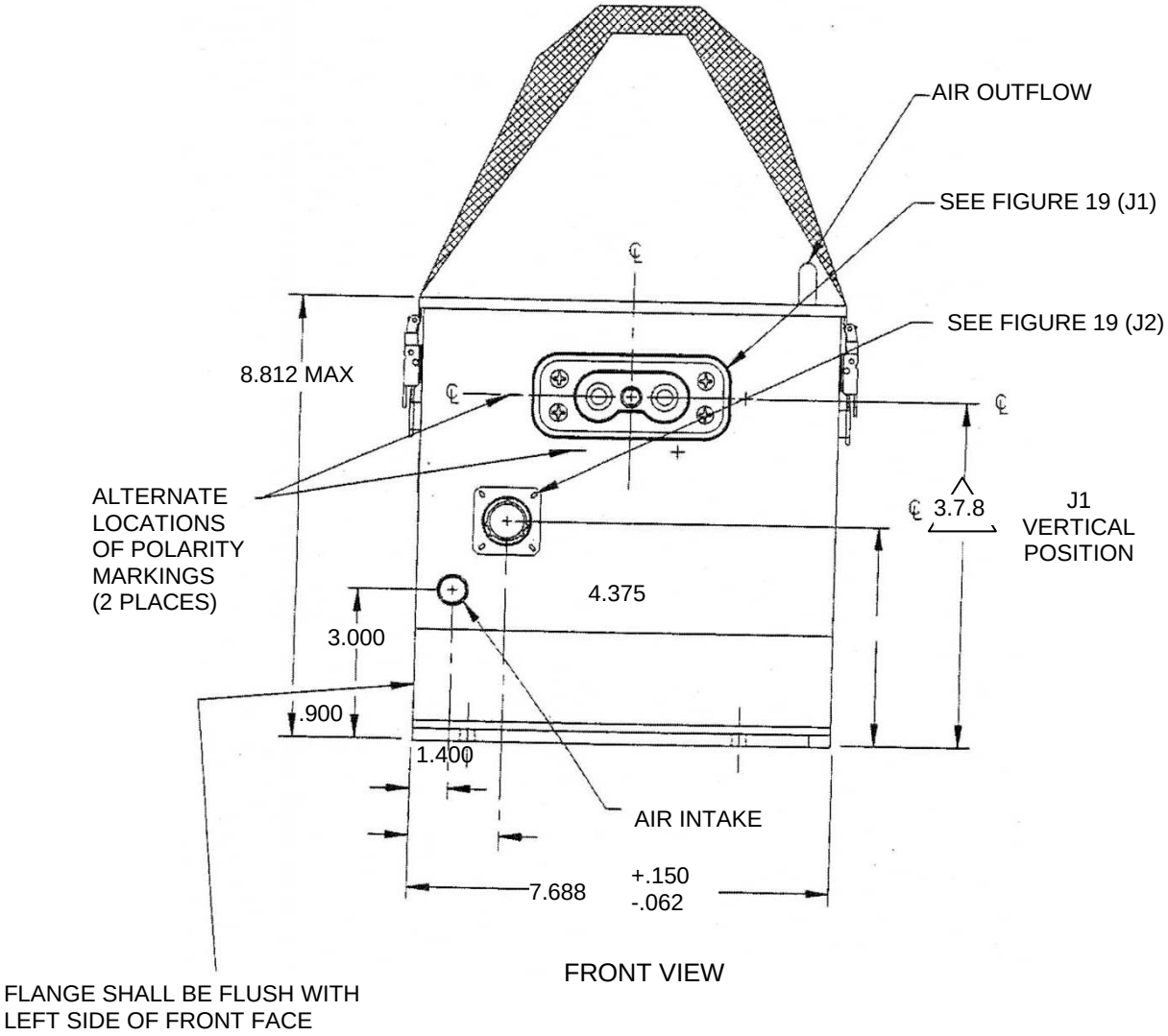


FIGURE 2. Dimensions and configuration for PIN M81757/15-2 - Continued.

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NOTE:

1. 3.7.8 DESIGNATES REQUIREMENTS PARAGRAPH 3.7.8 SPECIFIED HEREIN.

FIGURE 3. Dimensions and configuration for PIN M81757/15-3.

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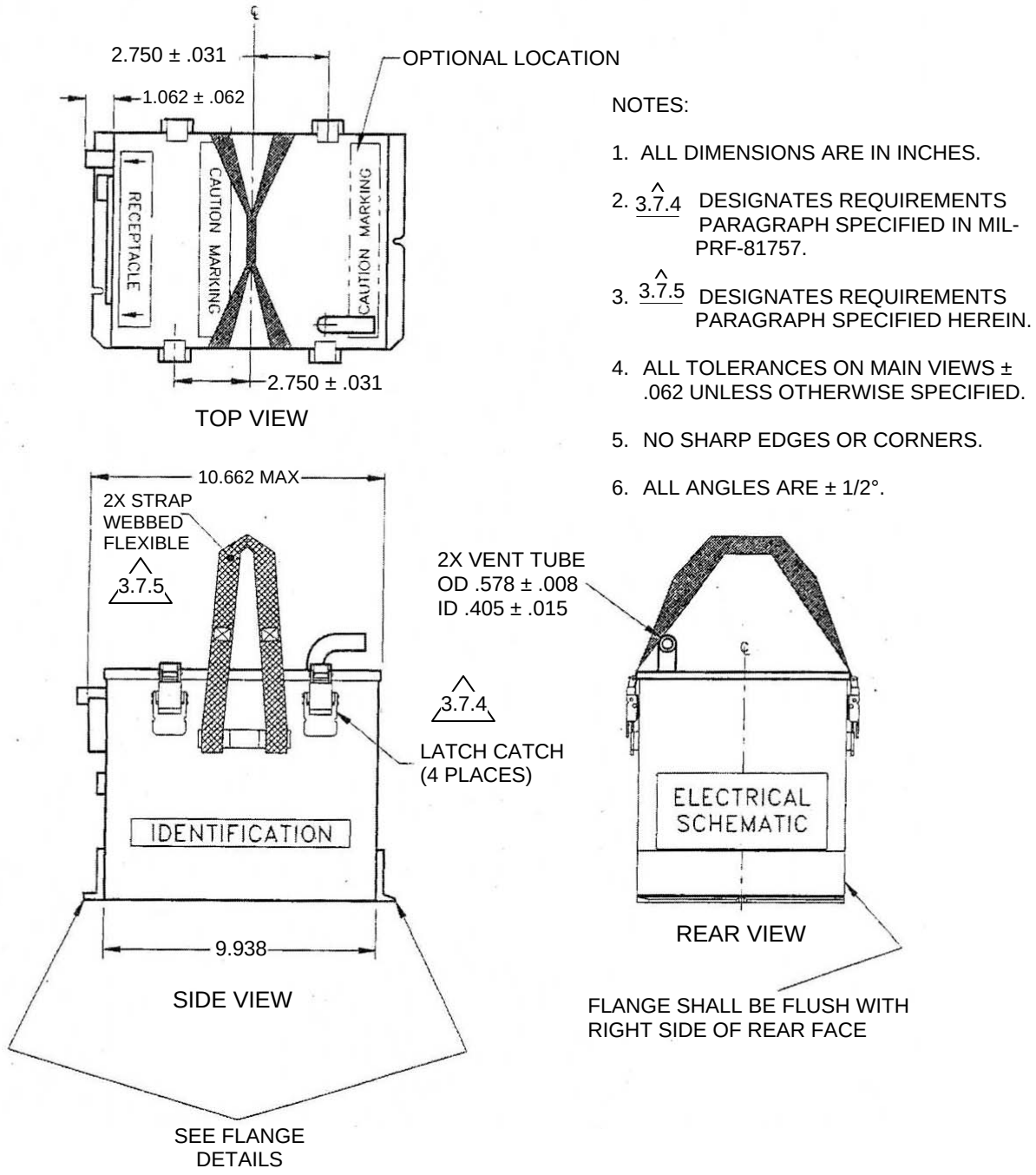
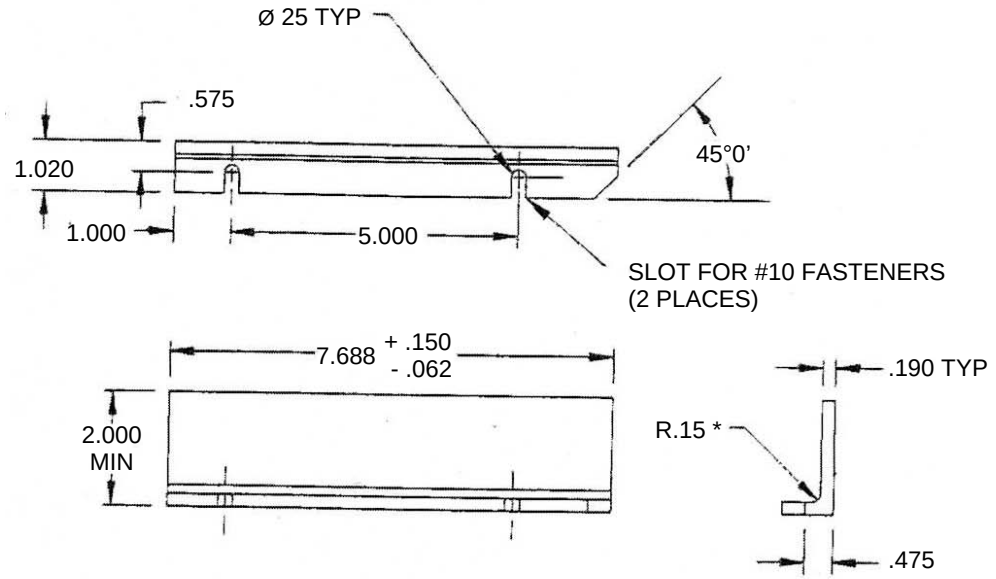
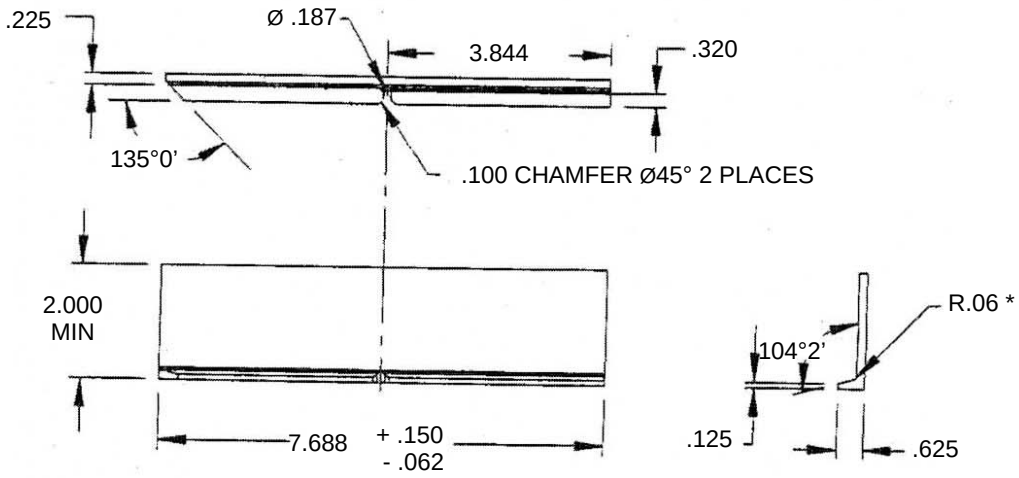


FIGURE 3. Dimensions and configuration for PIN M81757/15-3 - Continued.

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FRONT FLANGE DETAIL



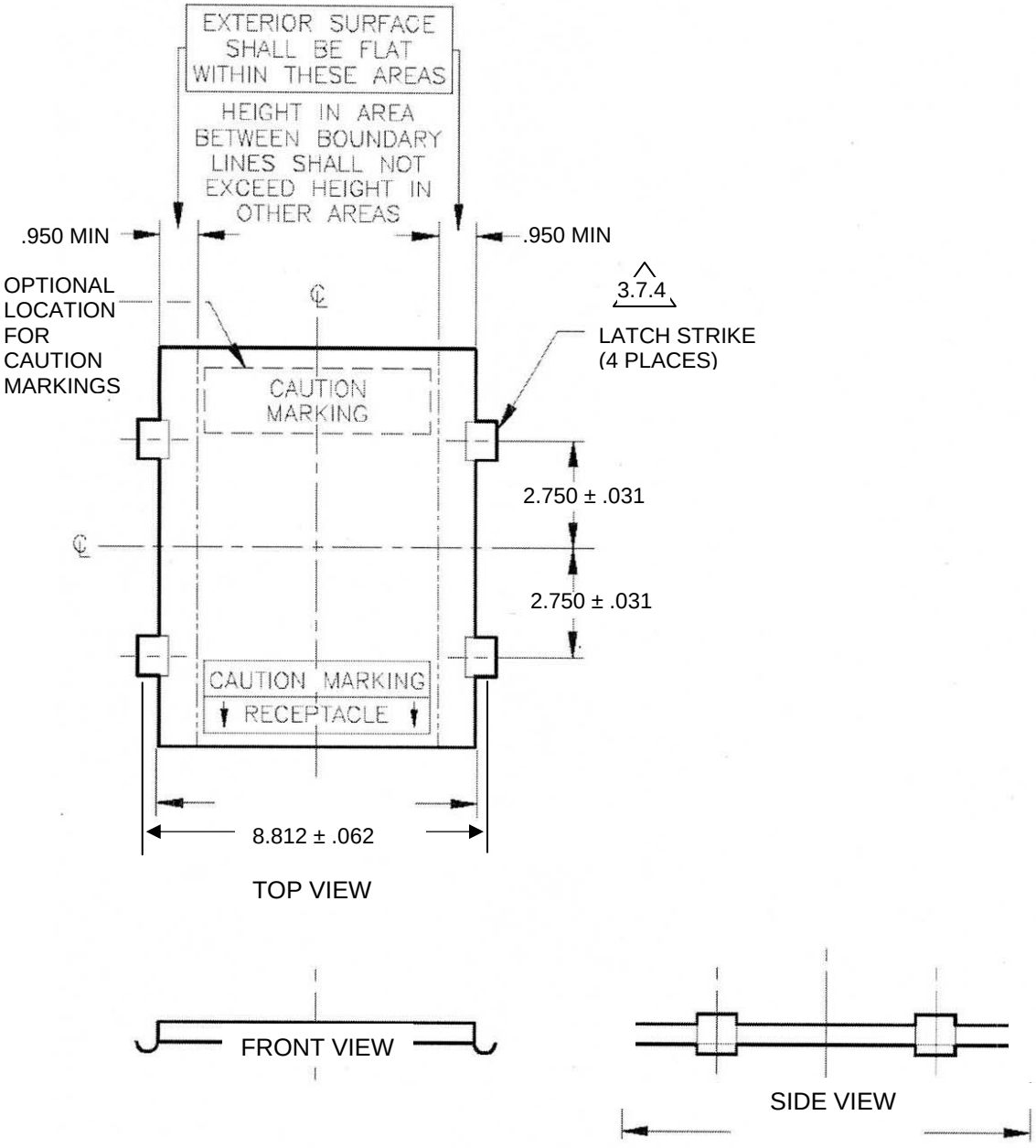
REAR FLANGE DETAIL

NOTES:

1. ALL TOLERANCES ON THIS FIGURE XXX ± .010 UNLESS OTHERWISE SPECIFIED.
2. NO SHARP EDGES OR CORNERS.
3. RADIUS VALUES DESIGNATED BY " * " ARE MAXIMUM TOLERANCES WITH A MINIMUM TOLERANCE OF 0, THUS A 90° ANGLE IS ACCEPTABLE.

FIGURE 3. Dimensions and configuration for PIN M81757/15-3 - Continued.

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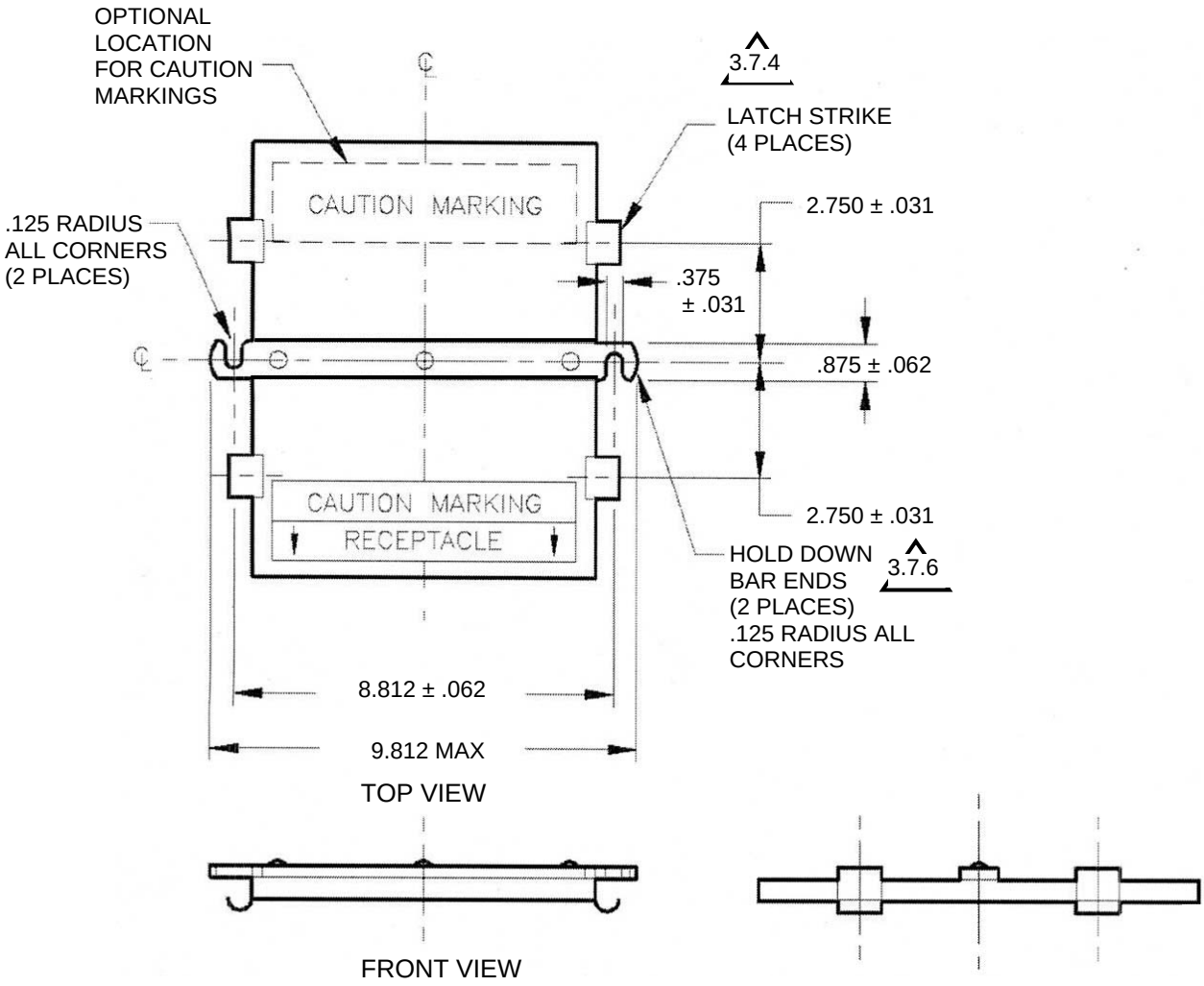


NOTES:

1. ALL DIMENSIONS ARE IN INCHES.
2. 3.7.4 DESIGNATES REQUIREMENTS PARAGRAPH IN MIL-PRF-81757.

FIGURE 4. Cover for PIN M81757/15-1.

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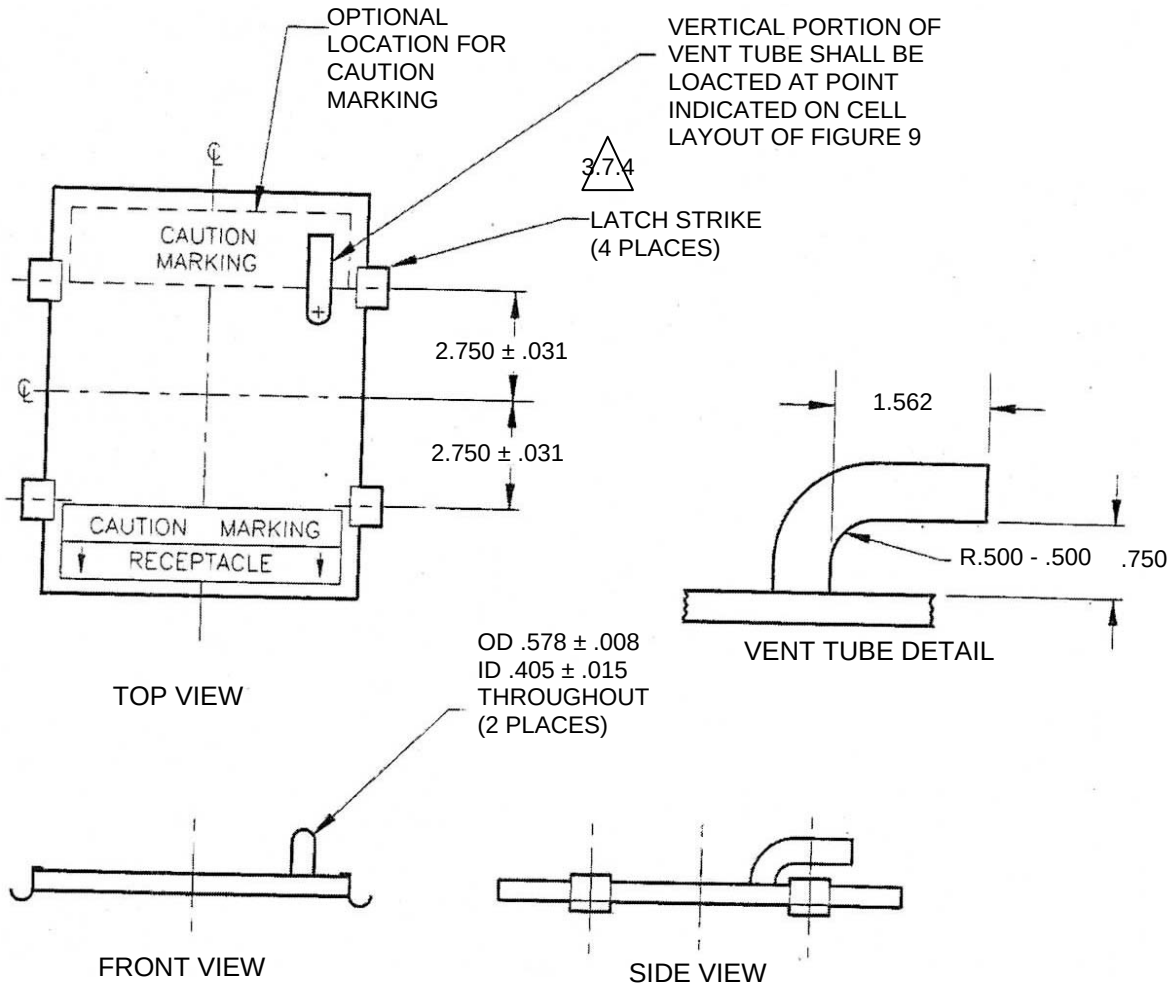


NOTES:

1. ALL DIMENSIONS ARE IN INCHES.
2. 3.7.4 DESIGNATES REQUIREMENTS PARAGRAPH SPECIFIED IN MIL-PRF-81757.
3. 3.7.6 DESIGNATED REQUIREMENTS PARAGRAPH SPECIFIED IN MIL-PRF-81757.

FIGURE 5. Cover for PIN M81757/15-2.

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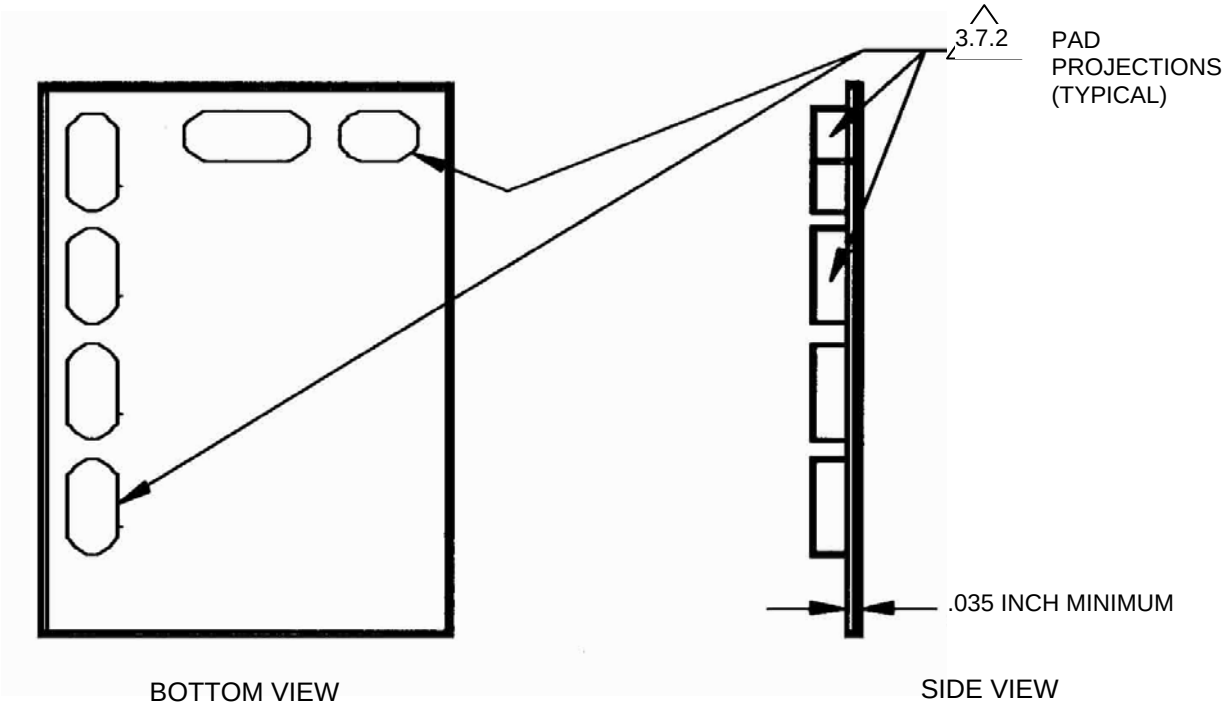


NOTES:

1. ALL DIMENSIONS ARE IN INCHES.
2. △ DESIGNATES REQUIREMENTS PARAGRAPH SPECIFIED IN MIL-PRF-81757.
3. ALL TOLERANCES ± .062 UNLESS OTHERWISE SPECIFIED.
4. NO SHARP EDGES OR CORNERS.
   3.7.4

FIGURE 6. Cover for PIN M81757/15-3.

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NOTES:

1. THE PAD SHALL COMPLY WITH 3.4, 3.7.2, 3.7.2.1, 3.7.2.2, AND 3.7.3.
2. 3.7.2 DESIGNATES REQUIREMENTS PARAGRAPH HEREIN.

FIGURE 7. Cover gasket/cell hold-down pad for M81757/15-X.

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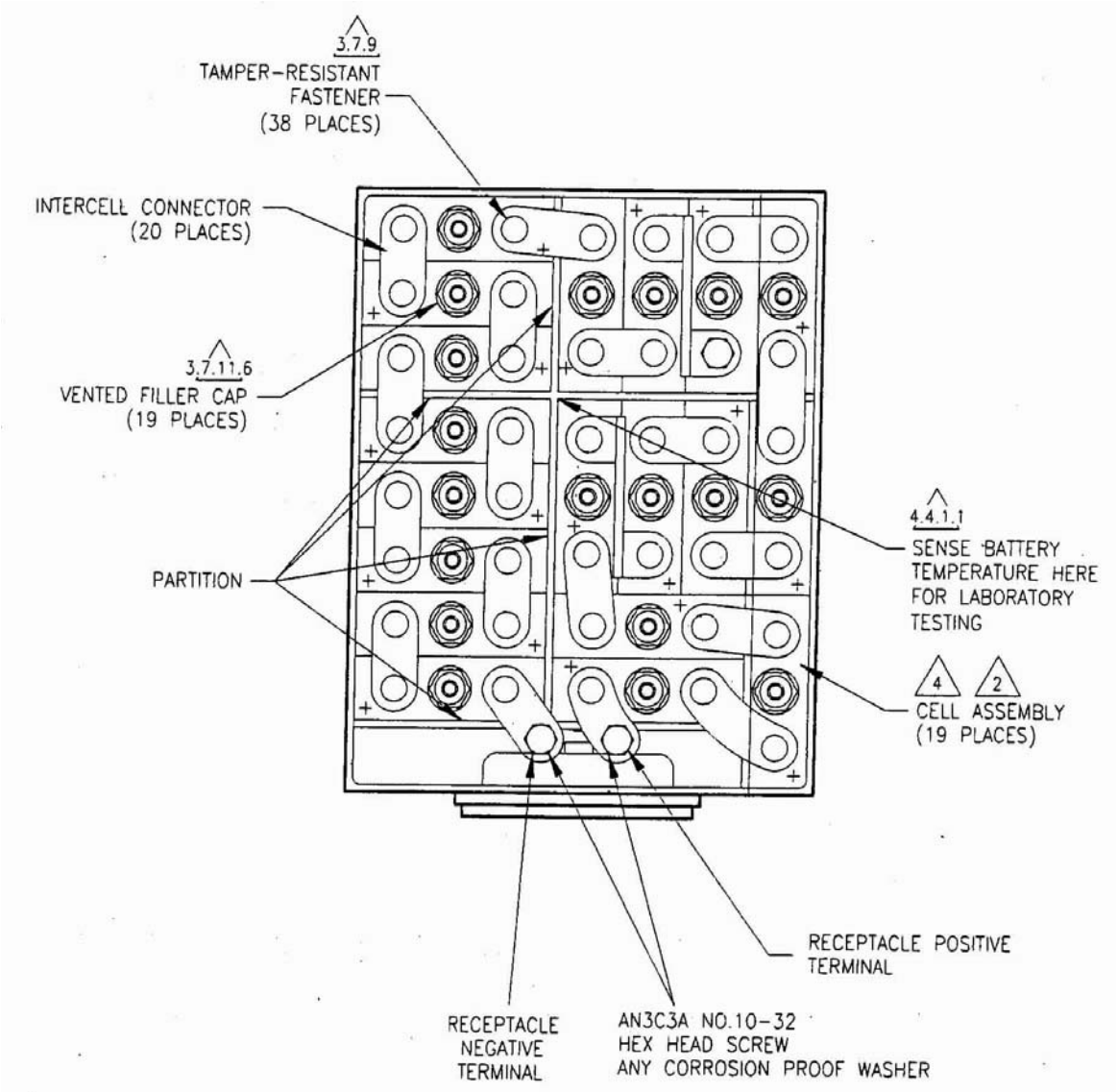
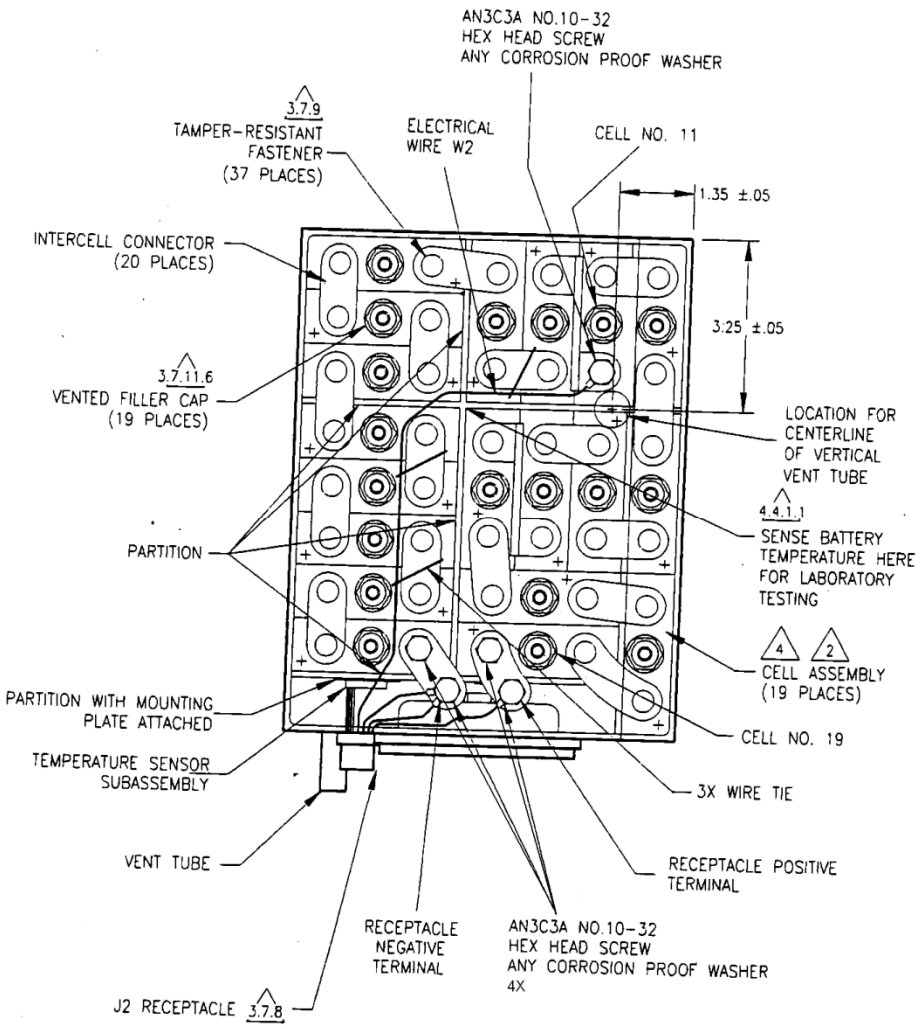


FIGURE 8. Layout of interior components for PINs M81757/15-1 and M81757/15-2.

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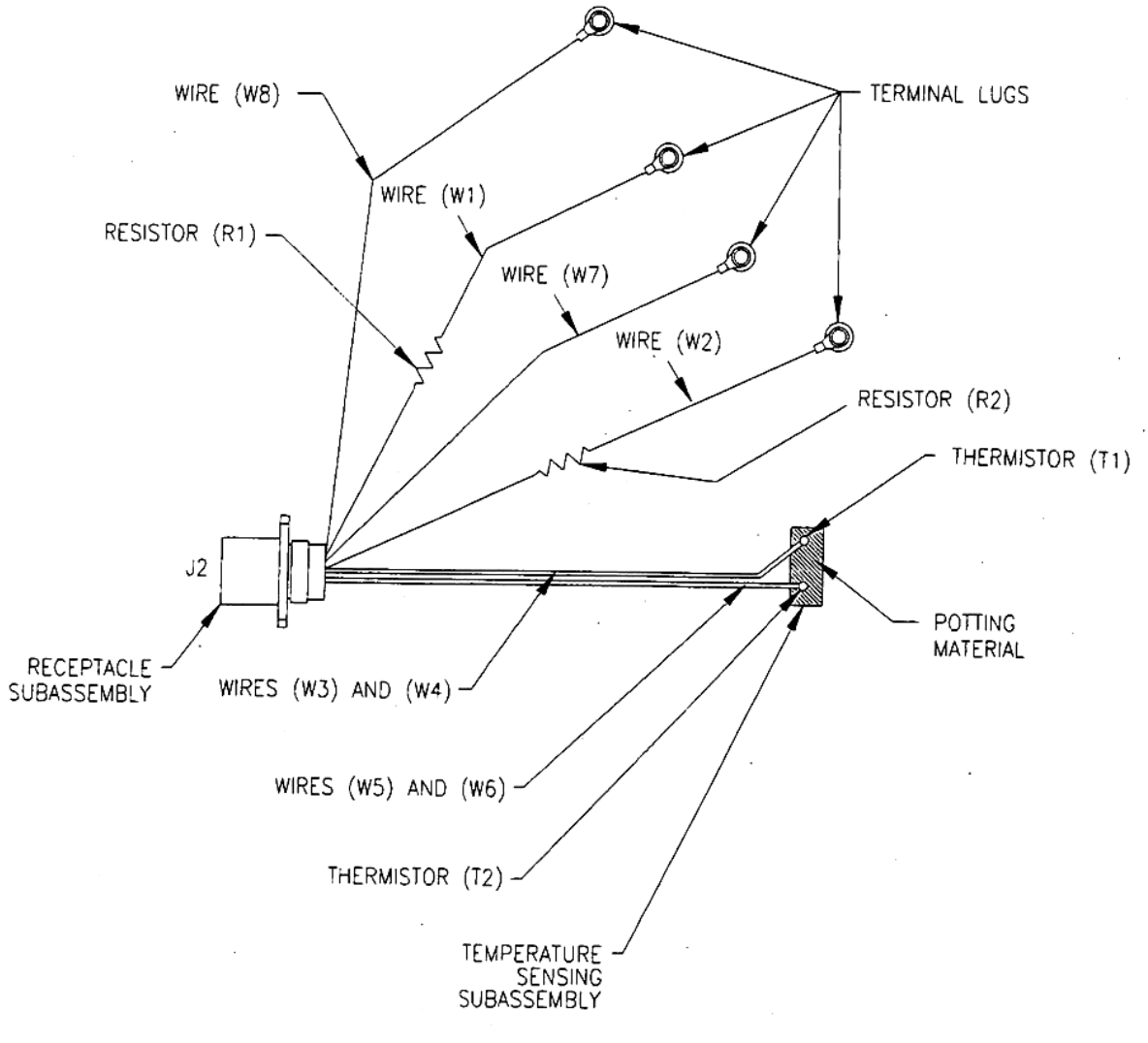
FRONT

NOTES:

1. 2 DESIGNATES REQUIREMENTS PARAGRAPH HEREIN.
2. 4 DESIGNATES REQUIREMENTS PARAGRAPH HEREIN.
3. 3.7.8 DESIGNATES REQUIREMENTS PARAGRAPH HEREIN.
4. 3.7.9 DESIGNATES REQUIREMENTS PARAGRAPH SPECIFIED IN MIL-PRF-81757.
5. 3.7.11.6 DESIGNATES REQUIREMENTS PARAGRAPH HEREIN.
6. 4.4.1.1 DESIGNATES REQUIREMENTS PARAGRAPH HEREIN.
7. MOUNTING FLANGES NOT SHOWN.

FIGURE 9. Layout of interior components for PIN M81757/15-3.

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NOTE:
1. TEMPERATURE SENSORING SUBASSEMBLY MUST BE FLUSH WITH PARTITION (ADJACENT TO THE CELLS).

FIGURE 10. PIN M81757/15-4 harness assembly for PIN M81757/15-3 battery.

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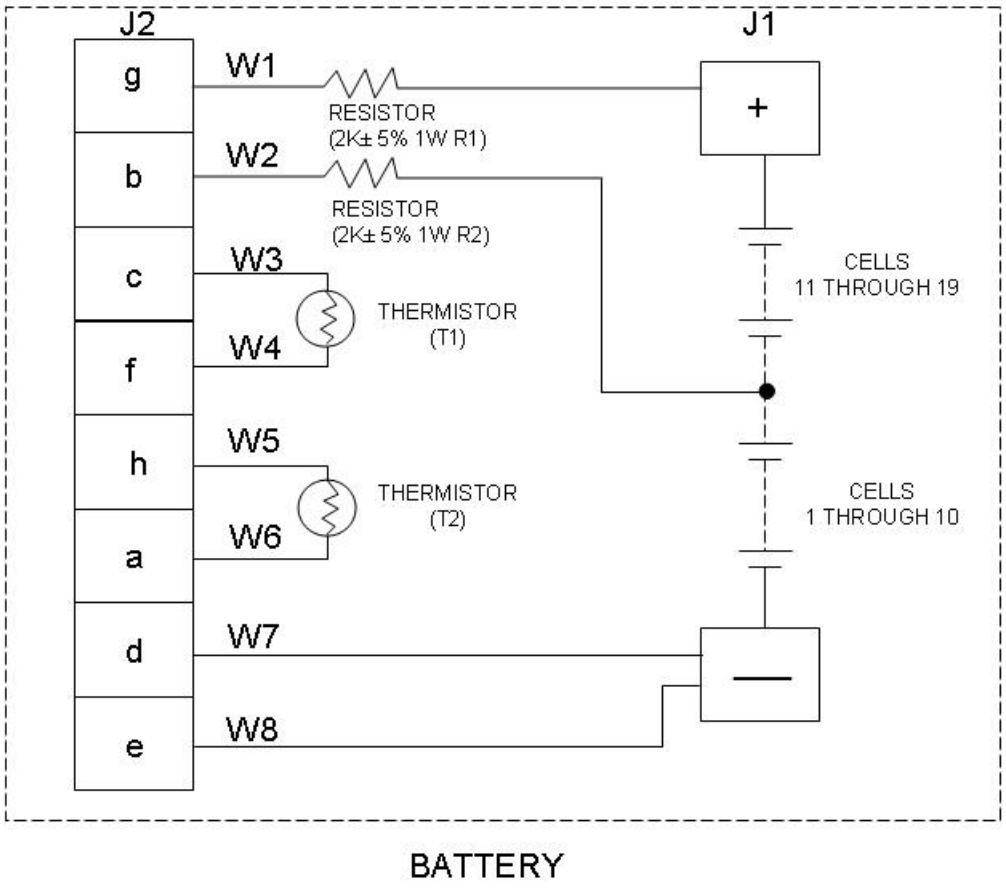


FIGURE 11. PIN M81757/15-4 harness assembly schematic and pin assignments for PIN M81757/15-3 battery.

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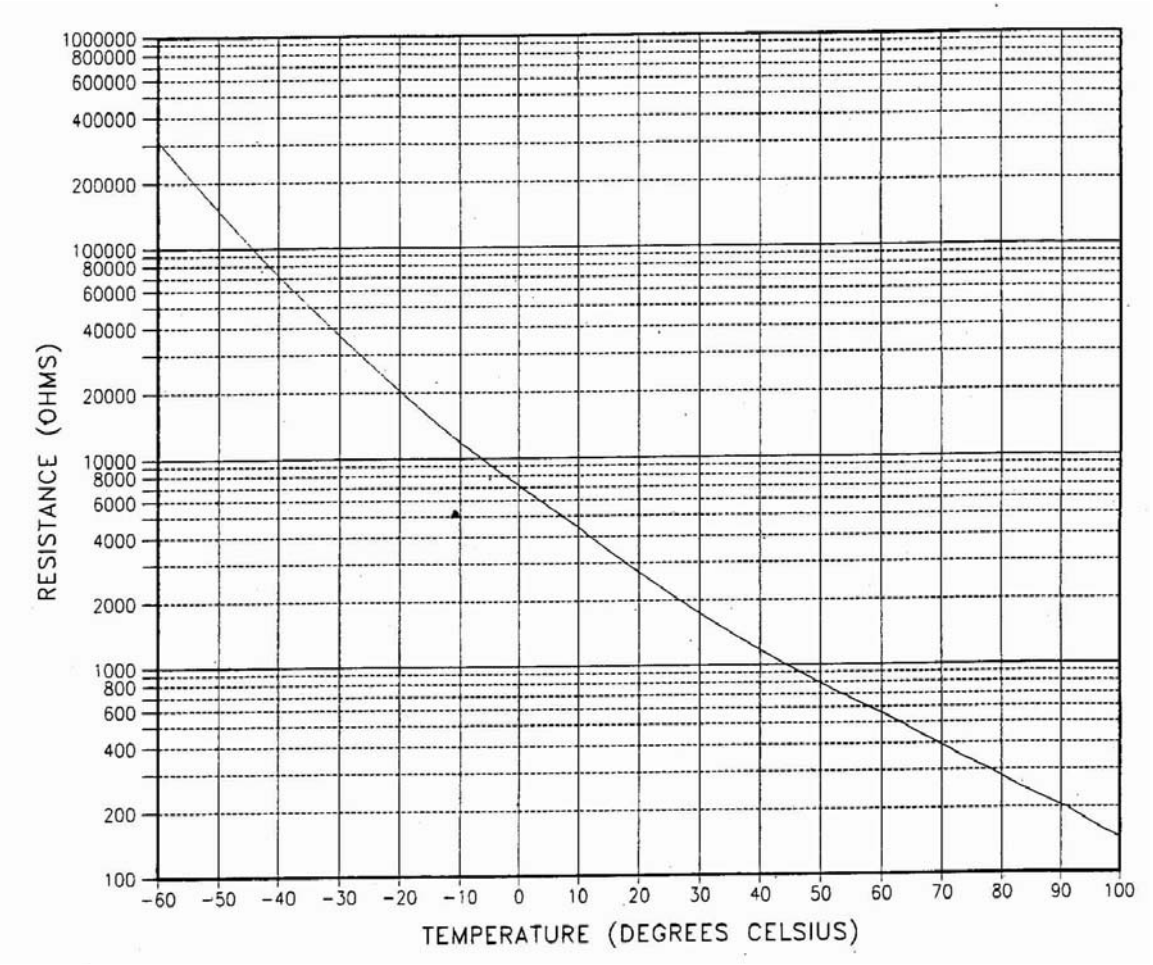


FIGURE 12. Resistance output for PIN M81757/15-3 temperature sensor devices.

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NOMENCLATURE:	BATTERY, STORAGE, AIRCRAFT, NICKEL-CADMIUM, HIGH RATE, LOW-MAINTENANCE, VENTED, LIMITED-REPAIR
MILITARY PIN:	M81757/15-X
NATIONAL STOCK NUMBER:	6140-**-***-****
NOMINAL VOLTAGE:	24 VOLTS
MAXIMUM WEIGHT:	***
INITIAL CAPACITY:	27.5 AH (25 AMPS/66 minutes/77 °F/18 V)
MINIMUM CAPACITY:	25 AH (25 AMPS/1 HR/77°F/18 V)
CONTRACT AND DELIVERY ORDER NUMBER:	***
MANUFACTURED BY:	***
MANUFACTURER'S CAGE:	***
MANUFACTURER'S SERIAL NUMBER:	***
DATE/LOT CODE:	***
DATE FIRST PLACED IN SERVICE:	

FIGURE 13. Marking for M81757/15-1 and M81757/15-2 batteries.

NOMENCLATURE:	BATTERY, STORAGE, AIRCRAFT, NICKEL-CADMIUM, HIGH RATE, LOW-MAINTENANCE, VENTED, LIMITED-REPAIR
MILITARY PIN:	M81757/15-X
NATIONAL STOCK NUMBER:	6140-**-***-****
NOMINAL VOLTAGE:	24 VOLTS
MAXIMUM WEIGHT:	***
INITIAL CAPACITY:	29 AH (25 AMPS/70 minutes/77 °F/18 V)
MINIMUM CAPACITY:	25 AH (25 AMPS/1 HR/77°F/18 V)
CONTRACT AND DELIVERY ORDER NUMBER:	***
MANUFACTURED BY:	***
MANUFACTURER'S CAGE:	***
MANUFACTURER'S SERIAL NUMBER:	***
DATE/LOT CODE:	***
DATE FIRST PLACED IN SERVICE:	

FIGURE 14. Marking for M81757/15-3 batteries.

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	MINIMUM LETTERING HEIGHT
CAUTION	0.500 inch minimum lettering height.
This cover manufactured by CAGE ***** and shall be installed only on a M81757/15-X battery manufactured by same CAGE.	0.250 inch minimum lettering height.
The M81757/15-X battery shall be serviced only in an authorized nickel-cadmium battery shop. The cells shall not be removed.	0.250 inch minimum lettering height.
⇩ RECEPTACLE ⇩	0.500 inch minimum lettering height.

FIGURE 15. Marking for battery caution and receptacle.

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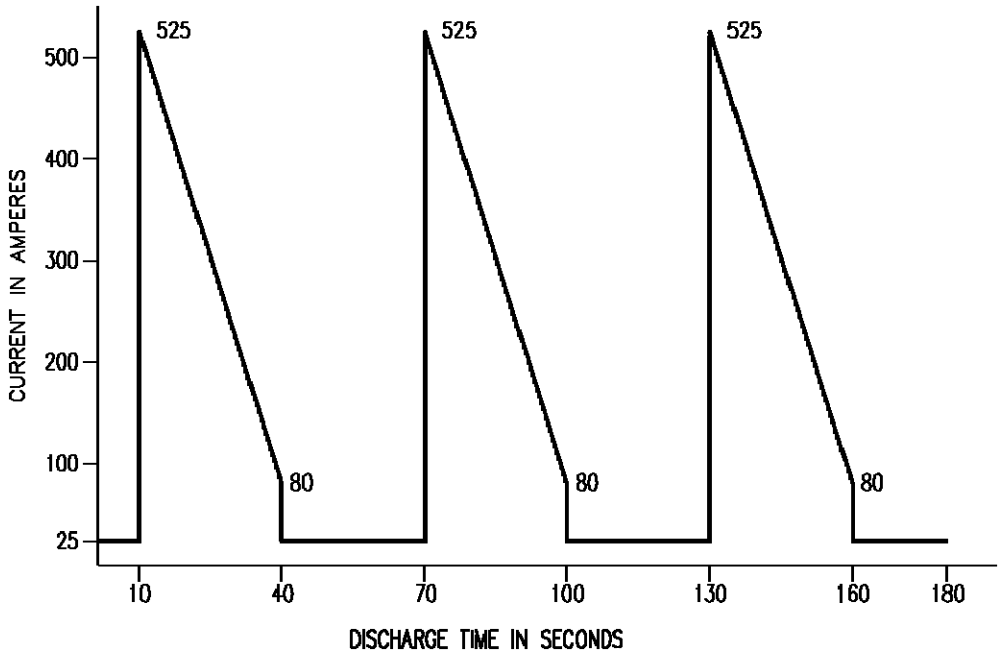


FIGURE 16. Slope discharge profile for PIN M81757/15-1.

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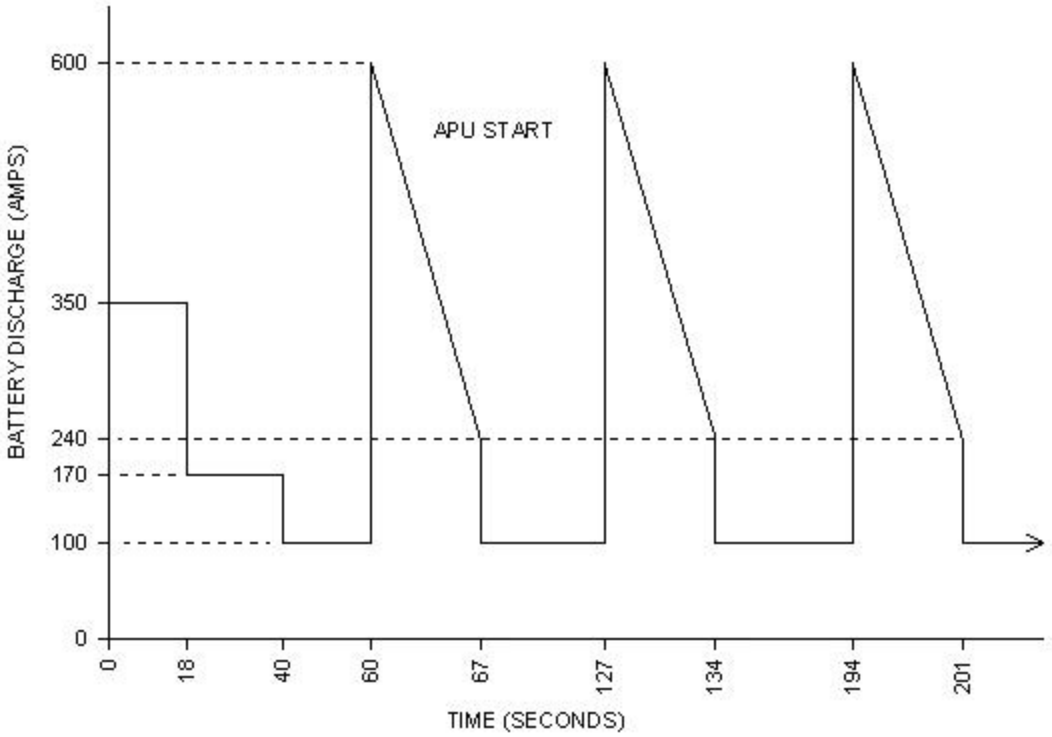


FIGURE 17. APU discharge profile for PIN M81757/15-3.

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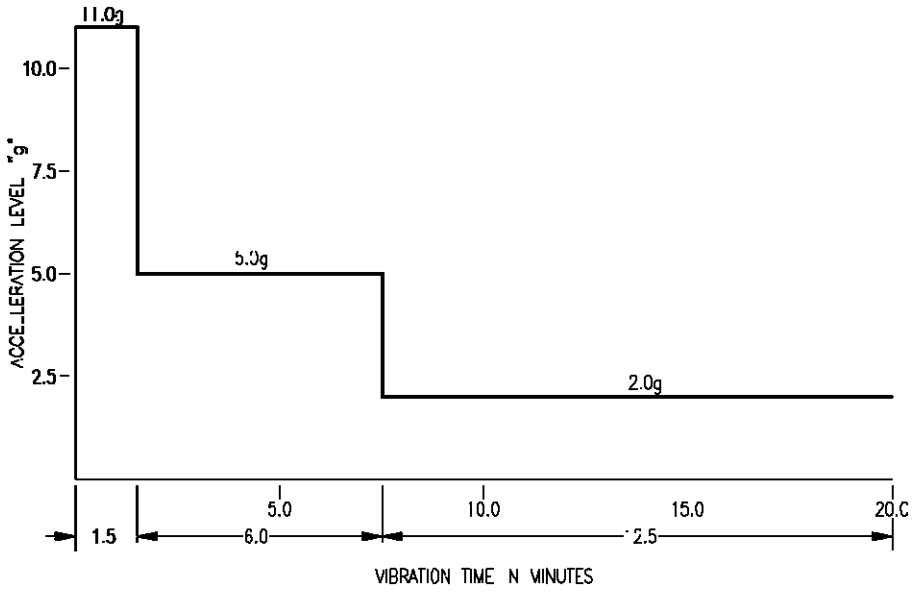


FIGURE 18. Low frequency (17 Hz) vibration for PIN M81757/15-1.

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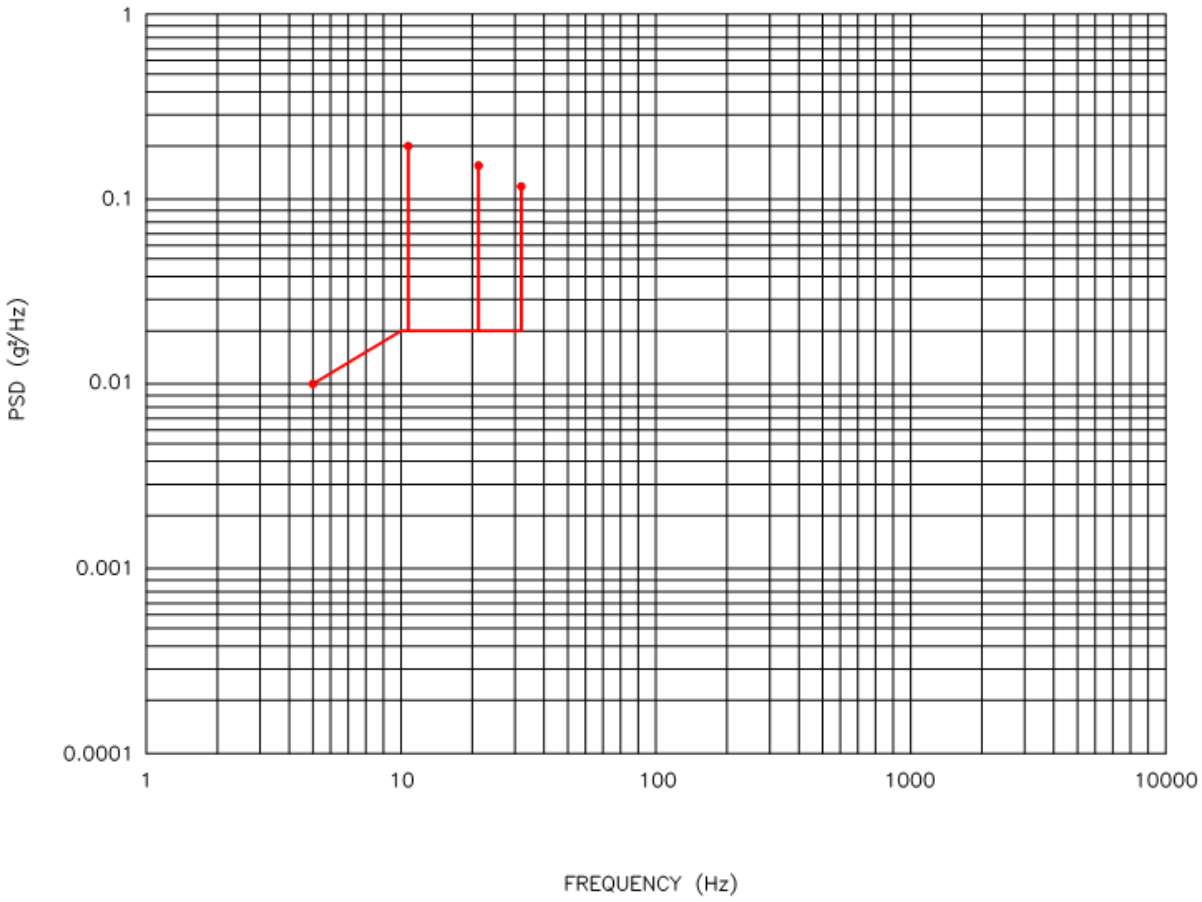


FIGURE 19. Gunfire vibration for PIN M81757/15-3.

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Table I. Broadband Random Input for Gunfire Qualification Tests

Frequency (Hz)	Amplitude (G^2/Hz)
5	0.01
10	0.05
100	0.05
200	0.03
300	0.05
2000	0.02

Table II. Narrow-band Random Input for Gunfire Qualification Tests

Frequency (Hz)	Amplitude (G^2/Hz)
11	0.20
22	0.18
33	0.12

Table III. Discrete Frequency Sinusoidal Excitation

Frequency (Hz)	Amplitude (G-peak)
19.1	1.0

FIGURE 19. Gunfire vibration for PIN M81757/15-3 - Continued.

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Applications	M81757/15-1	M81757/15-2	M81757/15-3
Applicable figures of MIL-PRF-81757/15	Figures 1, 4, 7, 8, 13, 15, 16, 18, and 20	Figures 2, 5, 7, 8, 13, 15, and 20	Figures 3, 6, 7, 9 through 12, 14, 15, 17, 19, and 20
Weight (lbs)	53.5 ± 1.5	52.0 maximum	56.0 maximum
Vented filler cap for cells	M81757/2-2	M81757/2-8	M81757/2-2
Handles	None	None	2 flexible straps
Main power receptacle J1	MS3509-32 (preferred) or MS3509-28	MS3509-32 (preferred) or MS3509-28	MS3509-32 (preferred) or MS3509-28
Auxiliary receptacle J2	None	None	D38999/ 20KC98SN
Battery mounting	No hold-down bar	Has hold-down bar	Special flanges
Special test/requirements	Low frequency vibration; Slope discharge	None	Gunfire vibration; unique cycling; APU discharge

FIGURE 20. Required characteristics for each PIN.

7.6 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

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CONCLUDING MATERIAL

Custodian:
Navy - AS

Preparing activity:
Navy - AS
(Project 6140-2010-013)

Agent:
Navy - SH

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.daps.dla.mil .

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